6CCSAPEA

Philosophy and Ethics of Artificial Intelligence

Revision Notes

Academic Year 2025/26 — Semester 1

Compiled on January 6, 2026

These notes cover philosophical concepts relevant to AI ethics: intelligence, consciousness, communication, ethics, algorithms, and societal impact.

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Module Overview

Introduction

Ethical issues relating to the use and deployment of current and future AI technologies are becoming more and more of a concern, and are regularly discussed and argued about in the media. The urgency with which these issues are discussed is in large part due to recent dramatic advances in machine learning (ML) approaches to AI, such as large language models, and the envisaged extent to which AI machines may be autonomous and act against our interests.

These ML advances are in turn driven by the increasing availability of big data and its use by machine learning algorithms, which are also exploited by social media companies to profile and predict our behaviours. We therefore see concerns about the deployment of autonomous cars and weapons, the possibility of super-intelligent machines and how their actions might inadvertently cause harm, and the problems of data bias and the use of data in targeted advertising.

This module reviews basic concepts in philosophy that are particularly relevant to how we should think about and address ethical concerns related to the AI algorithms of today and of the future.

Aims

On successful completion of this module, you will be able to demonstrate knowledge of:

- Basic philosophical work on the nature of **intelligence**, **consciousness**, **communication**, **society**, and **ethics**
- How this work relates to and impacts on developments in Artificial Intelligence
- The **ethical implications** of these developments

Learning Outcomes

By the end of this module, you will be able to:

PEA1: Discuss basic philosophical work on the nature of intelligence, consciousness, communication, society, and ethics

PEA2: Explain how this work relates to and impacts on developments in Artificial Intelligence

PEA3: Apply critical reasoning skills to challenging materials

PEA4: Engage in an informative way with issues of public concern regarding the dangers of Artificial Intelligence

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Course Structure

Week	Topic	Key Themes
1	Introduction	Module overview, philosophical foundations
2	Intelligence	What is intelligence? Turing Test, Chinese Room
3	Consciousness	Mind-body problem, qualia, machine consciousness
4	Reasoning & Communication	Logic, language, meaning, understanding
5	Ethics & Morality	Moral theories, ethical frameworks for AI
6	Algorithms	Algorithmic decision-making, bias, fairness
7	AI in Medicine & War	Applications, risks, ethical considerations
8	Superintelligence	Value loading, alignment, existential risk
9	AI & Human Society	Social impact, future of work, governance
10	Deceptive AI	(Not examined)
11	Review/Revision	Exam preparation

Key Philosophers and Concepts to Know

This module draws on work from many philosophers. Key figures likely to appear include:

- **Alan Turing** — Turing Test, computability, machine intelligence
- **John Searle** — Chinese Room argument, intentionality
- **Thomas Nagel** — "What is it like to be a bat?", consciousness
- **David Chalmers** — Hard problem of consciousness
- **Immanuel Kant** — Deontological ethics, categorical imperative
- **John Stuart Mill** — Utilitarianism, consequentialism
- **Nick Bostrom** — Superintelligence, existential risk

Note: This list will be expanded as we cover each week's content.

1 Week 1: Introduction

1.1 Topic Summary

Key Points(Learning Objectives).

By the end of this week, you will be able to:

1. Understand what **philosophy** is and its major branches.
2. Explain why philosophy is **relevant to AI**.
3. Describe how philosophical issues impact the **ethics of AI**.
4. Understand the **structure and assessment** of this module.

The Big Picture

This introductory lecture establishes why philosophical thinking matters for AI ethics. Philosophy provides the conceptual tools—about mind, knowledge, ethics, and logic—that we need to address questions like: Can machines think? Should autonomous systems be held responsible? How do we ensure AI benefits humanity?

The module is divided into two halves: (1) philosophical foundations (intelligence, consciousness, reasoning, ethics), and (2) applied AI ethics (algorithms, medicine/war, superintelligence, society).

What Examiners Like to Ask

- Define the major branches of philosophy (metaphysics, epistemology, ethics, logic).
- Explain Descartes' argument for mind-body dualism and its implications.
- How does determinism challenge traditional notions of moral responsibility?
- Demonstrate how logic can be used to analyse ethical arguments (Socratic method).
- Why are these philosophical concepts relevant to AI ethics?

1.2 Overview + Core Concepts + Extended Theory

Roadmap

1. What is philosophy?
2. Major branches of philosophy
3. Example 1: Descartes and mind-body dualism
4. Example 2: Free will, determinism, and responsibility
5. Example 3: Logic and ethical argument (Socratic method)
6. Relevance to AI ethics
7. Module structure overview

1.2.1 What is Philosophy?

Definition(Philosophy).

Philosophy (literally "love of wisdom") is an activity people undertake when they seek to understand **fundamental truths** about:

- Themselves (What is the self? What is consciousness?)
- The world (What exists? What is real?)
- Their relationships to the world and each other (What is right? How should we live?)

Philosophy addresses life's most basic questions:

- Why are we here? Why is there something rather than nothing?
- What is knowledge? How do we know what we know?
- What is consciousness? What is the self?
- What is the good life? What are morally right actions?

1.2.2 Major Branches of Philosophy

Branch	Focus	Key Questions
Metaphysics	Nature of reality, existence, being	Is there a God? What is truth? What is a person? Is the world composed of matter only? Do people have minds?
Epistemology	Nature of knowledge	What is knowledge? How do we know anything? How can we justify beliefs?
Ethics	What we ought to do	What is good? What makes actions right? Is morality objective or subjective?
Logic	Structure of arguments	What constitutes good reasoning? How do we evaluate arguments?

Exam Tip(Exam Relevance).

These four branches recur throughout the module:

- **Metaphysics** → Can machines have minds? (Weeks 2–3)
- **Epistemology** → Can machines know things? (Week 4)
- **Ethics** → What should AI do? (Weeks 5–9)
- **Logic** → How do we reason about ethics? (Week 4)

1.2.3 Example 1: Descartes and Mind-Body Dualism

Key Argument(Descartes' Cogito Argument (1637)).

"I think, therefore I am" (*Cogito ergo sum*)

The argument:

1. The very act of **thinking** proves the existence of the **self** (the "I" doing the thinking).
2. Everything else—the external world, our bodies—could be an illusion ("the Matrix").
3. We can be **absolutely certain** of the self, but **not certain** of the physical world.
4. Therefore, the **mind** (certain) must be fundamentally **different** from the **body** (uncertain).

Conclusion: Mind and body are separate substances (**dualism**).

Definition(Mind-Body Dualism).

The view that **mind** and **body** are fundamentally different kinds of things:

- Mind: non-physical, mental, private, certain
- Body: physical, material, public, uncertain

Example(Impact on Western Medicine).

Cartesian dualism dominated Western thought for centuries:

- Traditional view: Mind and body are **separate**, so mental processes **cannot affect** physical health.
- "Positive thinking" affecting health was **dismissed**.
- Modern view: Mind and body are **interconnected**. Mental states (stress, optimism) **do affect** physical health.
- Eastern medicine traditions (Buddhist, Ayurvedic) always recognised this connection.

Exam Tip(Relevance to AI).

The mind-body problem is central to AI:

- Can a **physical machine** have a **mind**?
- If mind and body are separate, could machines ever be conscious?
- If mind emerges from physical processes, perhaps machines *can* think.

We return to this in Weeks 2 (Intelligence) and 3 (Consciousness).

1.2.4 Example 2: Free Will, Determinism, and Responsibility

Definition(Determinism).

Determinism: All events, including human decisions, are determined by prior causes according to physical laws.

Implication: If physics is deterministic (A causes B causes C...), and mental processes are ultimately physical (neural activity), then:

- Every decision we make is the **effect** of a chain of causes.
- **Free will is an illusion**—there is no "self" that freely chooses independently of physics.

Key Argument(Implications for Justice).

If there is no free will, then when a criminal commits a crime:

- There is **no self** that "freely chose" to commit the crime.
- The crime was a **deterministic consequence** of prior causes (genetics, environment, neural processes).
- **Punishment** (retribution against a morally responsible self) may be unjustified.

Alternative justifications for penalties:

1. **Rehabilitation:** Change the criminal's decision-making so future decisions differ.
2. **Deterrence:** Signal to others that crime leads to penalty.
3. **Protection:** Protect society from the criminal (and vice versa).

Exam Tip(Relevance to AI).

These questions apply directly to autonomous AI:

- When an **autonomous car** kills a pedestrian, who is responsible?
- When a **lethal autonomous weapon** kills, who is accountable?
- Should we "punish" the AI? The designers? The users?
- How do we design penalties/incentives to prevent future harms?

We return to this in Week 6 (Algorithms) and Week 7 (AI in War).

1.2.5 Example 3: Logic and Ethical Argument

Definition(The Socratic Method).

Socrates used a method of **questioning** to expose **contradictions** in his interlocutors' arguments.

Key idea: By revealing inconsistencies, we can refine our beliefs and arrive at better-justified positions.

Example(Socrates vs. Sara: The Abortion Debate).

Dialogue:

1. Sara: "Abortion is wrong."
2. Socrates: "Why?"
3. Sara: "Human life is sacred, and a fetus is human."
4. Socrates: "So?"
5. Sara: "If a life is sacred, it should not be terminated."
6. Socrates: "So you should be against the death penalty for murderers?"
7. Sara: "Err... no! I'm in favour."

Sara's implicit reasoning:

1. If X is human → X's life is sacred.
2. A fetus is human → A fetus's life is sacred.
3. If X's life is sacred → X should not be terminated.
4. A fetus's life is sacred → A fetus should not be terminated.

The contradiction: The same reasoning applies to murderers (who are also human), so Sara should oppose the death penalty—but she supports it!

Key Argument(Refining Knowledge Through Refutation).

Solution: Sara can refine her rule to resolve the contradiction:

Old rule: If X is human → X's life is sacred.

Refined rule: If X is human **and** X does not take the life of another human → X's life is sacred.

Now Sara can consistently:

- Oppose abortion (fetus is human and hasn't killed anyone).
- Support the death penalty (murderer has taken a life).

Key insight: Knowledge evolves through refutation—we refine rules to accommodate counterexamples.

Exam Tip(Relevance to AI).

- **AI systems may need to reason about ethics** using logic.
- **Argumentation** (exchange of argument and counter-argument) is a formal method for ethical reasoning.
- We may use **AI-assisted reasoning** to help humans make better moral decisions.

We return to this in Week 4 (Reasoning) and Week 5 (Ethics).

1.2.6 Module Structure Overview

Week	Topic	Key Questions
1	Introduction	What is philosophy? Why relevant to AI?
2	Intelligence	What is intelligence? Can machines be intelligent?
3	Consciousness	What is consciousness? Could AI be conscious?
4	Reasoning & Communication	Symbolic vs. ML paradigms; argumentation
5	Ethics & Morality	Ethical theories; implementing moral agents
6	Algorithms	Bias, accountability, responsibility
7	AI in Medicine & War	Lethal autonomous weapons; medical AI ethics
8	Superintelligence	AGI → superintelligence; value loading problem
9	AI & Human Society	Echo chambers; surveillance capitalism; moral progress

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1.2.7 Summary: Key Takeaways

Key Argument(Week 1 Key Points).

1. **Philosophy** addresses fundamental questions about reality, knowledge, ethics, and reasoning.
2. **Descartes' dualism** separated mind from body—this matters for whether machines can have minds.
3. **Determinism** challenges free will—this matters for assigning responsibility to AI systems.
4. **Logic and the Socratic method** help us analyse and refine ethical arguments.
5. These philosophical tools will be applied throughout the module to AI ethics questions.

Definition(Key Terms to Remember).

- **Metaphysics:** Study of the nature of reality and existence.
- **Epistemology:** Study of knowledge and how we know things.
- **Ethics:** Study of what we ought to do; right and wrong.
- **Logic:** Study of the structure and validity of arguments.
- **Dualism:** Mind and body are fundamentally different substances.
- **Determinism:** All events are caused by prior events according to physical laws.
- **Modus ponens:** If P then Q; P; therefore Q.
- **Refutation:** Revising beliefs when contradictions are discovered.

2 Week 2: Intelligence

2.1 Topic Summary

Key Points(Learning Objectives).

By the end of this week, you will be able to:

1. Define **intelligence** and explain why the definition matters.
2. Distinguish between **Narrow AI**, **AGI**, and **Superintelligence**.
3. Explain the **Turing Test** and analyse its interpretations.
4. Evaluate the **Chinese Room** argument and its replies.
5. Understand why intelligence is **substrate-neutral**.

The Big Picture

This week addresses the foundational question: *What is intelligence?* We explore definitions of intelligence, examine whether machines can be intelligent, and critically analyse two famous thought experiments—the Turing Test and the Chinese Room. These concepts are essential for understanding debates about AI capabilities, consciousness, and the possibility of AGI.

What Examiners Like to Ask

- Define intelligence (Tegmark's definition) and explain Moravec's paradox.
- Explain the four quadrants of AI research (Russell & Norvig).
- Describe the Turing Test and its interpretations (sufficient/necessary conditions).
- Present the Chinese Room argument and evaluate replies (Systems, Robot, Virtual Mind).
- What is substrate neutrality and why does it matter for AI?

2.2 Overview + Core Concepts + Extended Theory

Roadmap

1. What is intelligence?
2. Substrate neutrality: information and computation
3. Four approaches to building AI (Russell & Norvig)
4. Types of AI: Narrow AI, AGI, Superintelligence
5. The Turing Test
6. Analysis of the Turing Test
7. The Chinese Room argument
8. Replies to the Chinese Room

2.2.1 What is Intelligence?

Definition(Intelligence (Tegmark)).

"Intelligence is the ability to accomplish complex goals."

This definition:

- Subsumes other definitions (logic, understanding, self-awareness, learning).
- Allows for **comparison**: X is more intelligent than Y if X can accomplish all goals Y can, plus at least one more.
- Results in a **partial ordering** (some agents are incomparable).

Definition(Moravec's Paradox).

Tasks that seem **easy for humans** (image recognition, walking) are **hard for computers**.

Tasks that seem **hard for humans** (complex calculations, chess) are **easy for computers**.

Explanation: Evolution has dedicated massive neural "hardware" to sensorimotor tasks, making them feel effortless. Computers lack this specialized circuitry.

Exam Tip(Exam Relevance).

Moravec's paradox explains why early AI succeeded at "hard" tasks (chess) but struggled with "easy" ones (vision). Modern deep learning has begun to address this by learning sensorimotor representations from data.

2.2.2 Substrate Neutrality: Information and Computation

Key Argument(The Substrate Neutrality Thesis).

Intelligence is ultimately about **information** and **computation**, not about the physical material (substrate) that implements them.

Implication: There is no reason in principle why machines (silicon-based) cannot be as intelligent as humans (carbon-based).

Definition(Information).

Information storage requires physical components that can exist in multiple **stable states**.

Examples:

- Hard drive: magnetized points (two states $\rightarrow$ 1 bit)
- RAM: electron positions
- Fiber optics: strong/weak laser beams
- DNA: nucleotide sequences
- Brain: $\sim$10 GB electrically (neuron firing), $\sim$100 TB chemically (synaptic strengths)

Key point: Information can be stored in **any** physical medium with stable states.

Definition(Computation).

Computation transforms one information state into another by implementing a **function**.

Key result: NAND gates are **Turing-complete**—any computable function can be implemented by connecting NAND gates.

Implication: A **universal intelligent machine** could, given enough time and resources, accomplish any goal as well as any other intelligent entity.

2.2.3 Four Approaches to Building AI

Russell and Norvig's classic textbook identifies four paradigms:

	Human-Based	**Ideal Rationality**
Reasoning-Based	Think like humans	Think rationally
Behaviour-Based	Act like humans	Act rationally

Definition(Ideal Rational Agent).

A **perfectly rational agent**:

1. Models the environment E.
2. Considers actions $a_1, \ldots, a_n$ and their resulting states $E_1, \ldots, E_n$.
3. Computes **expected utility** for each action: $\mathbb{E}[U(a_i)] = P(E_i \mid a_i) \times U(E_i)$.
4. Chooses the action with highest expected utility.

In practice: Perfect rationality is infeasible. We use **bounded optimality**—optimal behaviour given resource constraints (time, memory, processing power).

Exam Tip(AIXI Framework).

AIXI (Legg & Hutter) provides a mathematically rigorous definition:

*"Intelligence measures an agent's ability to achieve goals in a **wide range of environments**."*

Key insight: The agent doesn't know its environment in advance—it must learn and act simultaneously.

Problem: The optimal AIXI agent is **not Turing-computable**. Practical AI must approximate this ideal.

2.2.4 Types of AI: Narrow AI, AGI, Superintelligence

Type	Definition	Examples
Narrow AI	Purpose-built for **specific tasks**; often superhuman in that domain	Deep Blue (chess), AlphaGo, GPT-3
AGI	Can accomplish **any goal** at least as well as humans; possesses common sense, learning, reasoning	Does not yet exist
Superintelligence	**Greatly exceeds** human cognitive performance in virtually all domains	Hypothetical future AI

Definition(Moravec's Landscape).

Imagine a landscape where:

- **Lowlands** = tasks computers find easy (arithmetic, chess)
- **Peaks** = tasks humans find easy (locomotion, social interaction, creativity)

AI capability is like rising water—it has flooded the lowlands and is approaching the peaks. The **critical sea level** is when machines can perform **AI design itself**.

Key Argument(The Singularity).

Once machines can **design better versions of themselves**:

- Recursive self-improvement begins.
- Machines have advantages over human researchers: speed, memory, data access.
- Result: rapid **intelligence explosion** → Superintelligence.

Survey estimates (Bostrom):

- 50% of researchers expect AGI by 2040.
- 10% expect Superintelligence within 2 years of AGI.
- 75% expect Superintelligence within 30 years of AGI.

2.2.5 The Turing Test

Definition(The Turing Test (1950)).

Original formulation (Imitation Game):

- An interrogator communicates (via text) with a human and a machine in separate rooms.
- The interrogator tries to determine which is which.
- The machine tries to fool the interrogator; the human tries to help identify the machine.

Turing's prediction: In 50 years, a computer could fool an average interrogator 30% of the time after 5 minutes.

Alternative formulation: If the judge cannot do better than 50/50, the machine **passes** the test.

Exam Tip(Historical Note).

Turing anticipated **machine learning**: "Instead of trying to produce a programme to simulate the adult mind, why not... simulate the child's? If... subjected to an appropriate course of education one would obtain the adult brain."

Descartes (1669) prefigured the test: machines could never "arrange speech in various ways, in order to reply appropriately to everything that may be said in its presence."

2.2.6 Analysis of the Turing Test

Definition(Logical Interpretations).

The Turing Test can be interpreted as providing:

1. **Logically sufficient conditions:**

$$\text{Pass TT} \Rightarrow \text{Intelligent}$$

(TT-passers are a **subset** of intelligent beings. Failing TT doesn't prove non-intelligence.)

2. **Logically necessary conditions:**

$$\text{Intelligent} \Rightarrow \text{Pass TT}$$

(Intelligent beings are a **subset** of TT-passers. Failing TT proves non-intelligence.)

3. **Both (biconditional):** Pass TT $\Leftrightarrow$ Intelligent. (Rarely claimed.)
4. **Probabilistic:** Passing TT provides **evidence** for intelligence (Turing's own view).

Example(Objections to the Turing Test).

1. **The Chauvinistic Objection** (attacks necessary interpretation):
   - Intelligent aliens might fail because they don't share human linguistic conventions.
   - An entity could achieve complex goals in diverse environments without passing TT.
2. **The Blockhead Objection** (attacks sufficient interpretation):
   - Imagine a creature with a **lookup table** containing responses to every possible input.
   - Blockhead passes TT but uses **brute force**—no real intelligence or understanding.
   - **Counter:** Is Blockhead logically/nomically possible? Is brute-force processing really non-intelligent?
3. **Too Hard:** Nothing without a human-like brain could pass.
4. **Too Narrow:** Success might come from tricks, not genuine cognition.
5. **Too Easy:** More demanding tests needed (e.g., **Lovelace Test**—creativity/originality).

Exam Tip(Exam Tip).

When discussing TT interpretations, use set-theoretic reasoning:

- Sufficient: {TT-passers} $\subseteq$ {Intelligent beings}
- Necessary: {Intelligent beings} $\subseteq$ {TT-passers}

Then apply this to analyse objections.

2.2.7 The Chinese Room Argument (Searle, 1980)

Key Argument(The Chinese Room).

Setup:

- A person who knows no Chinese is locked in a room.
- The room contains boxes of Chinese symbols (database) and a rulebook (program).
- People outside send in Chinese questions (input).
- The person follows rules to manipulate symbols and produces Chinese answers (output).
- The person passes the Turing Test for Chinese—but understands nothing.

Argument structure:

1. The symbol-manipulating program P is possible (at least conceivably).
2. P does not possess **understanding** $\rightarrow$ P is not intelligent.
3. Any digital computer C is equivalent to P in relevant respects.
4. **Conclusion:** No digital computer can be intelligent.

Definition(Key Distinction: Syntax vs. Semantics).

- **Syntax:** Formal manipulation of symbols according to rules.
- **Semantics:** Meaning—what symbols **represent**.

Searle's claim: Computers can only do syntax. Understanding requires semantics. Therefore, computers cannot truly understand—and understanding is necessary for intelligence.

Challenge: Is understanding really necessary for intelligence (per Tegmark's definition)?

2.2.8 Replies to the Chinese Room

Reply	Argument
Systems Reply	The **man** doesn't understand, but the **whole system** (man + database + rulebook) does. The man is just the CPU.
Virtual Mind Reply	The system creates a **virtual entity** (like Siri or a video game character) that understands, distinct from the physical components.
Robot Reply	Understanding requires **embodiment**—interacting with the world. A robot that sees, touches, and experiences could ground symbols in meaning.
Impossibility Reply (Dennett)	Such a program would require ∼100 billion lines of code. Running it by hand would take lifetimes. It's not **nomically possible**.

Definition(Nomic vs. Logical Possibility).

- **Logical possibility:** No inherent contradiction (can be conceived).
- **Nomic possibility:** Possible given the **laws of physics**.

Dennett argues the Chinese Room is **logically** possible but not **nomically** possible—so it's not a valid counterexample.

Exam Tip(Exam Tip).

The Chinese Room hinges on **Premise 2**: understanding is necessary for intelligence. If you accept Tegmark's goal-based definition, you can reject this premise—a system could be intelligent (achieve complex goals) without "true understanding."

2.2.9 Summary: Key Takeaways

Key Argument(Week 2 Key Points).

1. **Intelligence** = ability to accomplish complex goals (Tegmark).
2. **Moravec's paradox:** "Easy" human tasks are computationally hard.
3. **Substrate neutrality:** Intelligence depends on information/computation, not physical material.
4. **Four AI paradigms:** Think/act like humans vs. think/act rationally.
5. **Narrow AI** excels at specific tasks; **AGI** matches human breadth; **Superintelligence** exceeds humans in all domains.
6. **Turing Test:** Linguistic indistinguishability as a test for intelligence.
7. TT can be interpreted as sufficient, necessary, both, or probabilistic.
8. **Chinese Room:** Syntax alone doesn't yield understanding.
9. Replies: Systems, Virtual Mind, Robot (embodiment), Impossibility.

Definition(Key Terms to Remember).

- **Narrow AI / AGI / Superintelligence:** Levels of AI capability.
- **Substrate neutrality:** Intelligence is hardware-independent.
- **Turing-complete:** Can compute any computable function.
- **Expected utility:** $P(\text{outcome}) \times U(\text{outcome})$.
- **Bounded optimality:** Best behaviour given resource limits.
- **Turing Test:** Behavioural test for machine intelligence.
- **Logically sufficient/necessary:** Set-theoretic interpretations.
- **Chinese Room:** Thought experiment against computer understanding.
- **Syntax vs. Semantics:** Symbol manipulation vs. meaning.
- **Nomic possibility:** Possible given physical laws.
- **Embodiment:** Understanding requires physical interaction with the world.

3 Week 3: Consciousness

3.1 Topic Summary

Key Points(Learning Objectives).

By the end of this week, you will be able to:

1. Define **consciousness** and distinguish it from intelligence.
2. Explain the **Easy Problem** vs. the **Hard Problem** of consciousness.
3. Compare **dualist** and **physicalist** theories of consciousness.
4. Evaluate the **Philosophical Zombies** argument.
5. Describe **information-theoretic** theories (GWT, IIT) and their implications for AI.
6. Assess whether **AI could be conscious**.

The Big Picture

Consciousness—subjective experience, "what it is like" to be something—is one of the deepest mysteries in philosophy. This week explores whether machines could ever be conscious, which matters because: (1) consciousness may or may not be necessary for intelligence, (2) if AI is conscious, it may experience suffering and deserve moral consideration, (3) how we treat AI (as if conscious or not) will shape society.

What Examiners Like to Ask

- Define consciousness (Nagel's formulation). How does it differ from intelligence?
- Explain the Easy Problem vs. Hard Problem (Chalmers).
- Present the Philosophical Zombies argument. What does it try to show?
- Compare dualism and physicalism. What is functionalism?
- Describe Global Workspace Theory and Integrated Information Theory.
- Could AI be conscious? What criteria might we use?

3.2 Overview + Core Concepts + Extended Theory

Roadmap

1. What is consciousness?
2. Revisiting the Chinese Room: understanding vs. experience
3. The Easy Problem vs. the Hard Problem
4. Theories of consciousness: Dualism and Physicalism
5. The Philosophical Zombies argument
6. What is a conscious state? (Qualia, access consciousness)

7. Roles and functions of consciousness
8. Information-theoretic theories (GWT, IIT)
9. Could AI be conscious?

3.2.1 What is Consciousness?

Definition(Consciousness (Tegmark)).

Consciousness = subjective experience.

The "thing that goes away every night in deep sleep, and comes back when we wake up." It encompasses all our subjective feelings and experiences.

Definition(Nagel's Formulation (1974)).

*"Something is conscious if there is **something that it is like** to be that something."*

There is some subjective way the world seems or appears from the creature's mental/experiential point of view.

Example: "What is it like to be a bat?" — Bats use echolocation. Is there a subjective experience of echolocating? We can never fully know.

Exam Tip(Why Consciousness Matters for AI).

1. **Is consciousness necessary for intelligence?** Definitions of AGI/Superintelligence don't require it—but that doesn't mean it's not needed.
2. **Moral consideration:** If AI is conscious, it may experience pain/suffering → ethical obligations.
3. **Social impact:** If we *treat* AI as conscious (regardless of whether it is), this will profoundly change society.

3.2.2 Revisiting the Chinese Room: Understanding vs. Experience

Key Argument(Searle's Revised Conclusion (2010)).

*"The implementation of the computer program is not by itself sufficient for **consciousness** or **intentionality**."*

Intentionality = "aboutness"—the way mental states relate to things in the world.

Implication: A computer may *behave* as if it understands, but lack **real understanding** (conscious apprehension of meaning).

Definition(Functionalism).

The view that mental states count as being of a given type because of the **functional role** they play.

"If it looks like a duck, walks like a duck, and quacks like a duck—it is a duck."

Applied to consciousness: If system S_1 produces the same input/output pairs as a conscious system S_2, then S_1 is conscious (regardless of what S_1 is made of).

Searle rejects functionalism: Behaviour isn't enough; genuine understanding requires conscious experience.

Example(Dennett's "Competence Without Comprehension").

Modern NLP systems (GPT-3, etc.) respond to questions based on learned contexts—"the meaning of a word is the company it keeps."

They exhibit **competent behaviour** without (arguably) any **conscious comprehension**.

Question: If a machine can do everything that understanding enables, why insist on an "inner feeling" of understanding?

3.2.3 The Easy Problem vs. the Hard Problem (Chalmers)

Definition(The Easy Problem).

How does the brain work in all its functions?

- How does it interpret sensory inputs?
- How does it produce behaviour and language?
- How do brain processes support cognition?

These are **extremely difficult** but are problems of **mechanism**—they don't require explaining subjective experience.

Definition(The Hard Problem).

How does physical matter give rise to **feelings** and **experience**—the "what it seems like from the inside"?

When you drive, you experience colours, sounds, emotions. Does a self-driving car experience **anything at all**?

Both systems process information and output motor commands. But **subjective experience** seems logically separate—is it optional? What causes it?

The explanatory gap: We have no satisfactory explanation of how consciousness arises from the physical.

Exam Tip(Exam Tip).

The Easy/Hard distinction is central. Easy problems are about *function*; the Hard Problem is about *experience*. Information-theoretic theories (GWT, IIT) address the Easy Problem but arguably don't fully solve the Hard Problem.

3.2.4 Theories of Consciousness: Dualism and Physicalism

Theory	Core Claim
Substance Dualism	Two kinds of substance exist: physical matter and non-physical consciousness (Descartes).
Property Dualism	One substance (matter), but two kinds of properties: physical and conscious. Conscious properties are not reducible to physical properties.
Physicalism	Everything is physical. Consciousness can (in principle) be explained in terms of physical processes.
Functionalism	(A form of physicalism) Mental states are defined by their functional role—what they do, not what they're made of.
Neutral Monism	Underlying reality is neither physical nor mental—both are aspects of something more fundamental (Bertrand Russell).

3.2.5 The Philosophical Zombies Argument

Definition(Philosophical Zombie).

A **philosophical zombie** is an exact physical duplicate of a person that:

- Behaves identically to a conscious person.
- Has identical brain states, neuron firings, etc.
- But has **no experiential consciousness**—there is nothing it is like to be a zombie.

Key Argument(The Zombie Argument (Chalmers)).

Premises:

1. Zombies are **conceivable** (we can imagine them).
2. If conceivable, then **possible** (at least logically).
3. If consciousness were a physical property, a physical duplicate would *necessarily* be conscious.
4. But zombies (physical duplicates without consciousness) are possible.

Conclusion: Consciousness is **not** a physical property. Therefore, **property dualism** is true.

Example(Objections to the Zombie Argument).

1. Is conceivability → possibility valid?

- You can *imagine* a 747 flying backwards, but the more you know about aerodynamics, the harder it becomes.
- The more we understand brain processes, the less conceivable zombies may become.

2. Can we really conceive of zombies?

- If consciousness is tightly linked to brain function, perhaps we *can't* coherently conceive of a physical duplicate without experience.

Even if you reject the zombie argument, the Hard Problem remains: How do we explain subjective feelings in terms of physical relations?

3.2.6 What is a Conscious State?

Conscious states have several distinct but interrelated meanings:

Type	Definition	Could AI have it?
Meta-awareness	A mental state one is **aware of being in**. To desire coffee *and* be aware you have that desire.	In principle, yes.
Qualia	Raw sensory feels—the taste of wine, the redness of red. **Phenomenal properties.**	Hard Problem applies.
Access Consciousness	A state is conscious if it **relates to, is available to, and interacts with** other states. Information is accessible for use and guidance.	In principle, yes.

Definition(Qualia and Phenomenal Structure).

Qualia (singular: quale) = the qualitative, experiential properties of conscious states.

Phenomenal structure = the overall structure of experience—not just sensory qualia but also spatial, temporal, and conceptual organization of our experience of the world and ourselves.

Problem: Is it possible to give a fully descriptive account of phenomenal experience from the outside?

Example(Inverted Qualia).

How do I know that what *you* experience when you see red is the same as what *I* experience?

We have circumstantial evidence (similar neural structures, similar reactions), but we can never be 100% certain.

3.2.7 Roles and Functions of Consciousness

Key Argument(Is Consciousness Epiphenomenal?).

Epiphenomenalism: Consciousness is a byproduct that doesn't play any causal role—like steam from a train engine.

Evidence: Experiments show that decisions are often initiated *before* we consciously "decide"—the experience of deciding may be post-hoc reporting.

Counter: Even if initially epiphenomenal, consciousness may have evolved a signalling function (like steam signalling a train's approach to a town).

Definition(Functional Roles of Consciousness).

1. **Flexibility and control:**
   - Meta-awareness increases flexibility (vs. fast but rigid unconscious processes).
   - Especially important for **novel situations**—a hallmark of intelligence.
2. **Social coordination:**
   - Awareness of our own mental states helps us understand others' mental states (**Theory of Mind**).
   - Enables cooperation, communication, joint reasoning.
3. **Unified, integrated representation:**
   - Conscious experience integrates information from multiple sensory channels + memory + background knowledge.
   - Allows more sophisticated and flexible responses.
4. **Global informational access:**
   - Conscious information is available to diverse mental subsystems.
   - Widens the sphere of influence of information.

Exam Tip(Interoception and Evolution).

Interoception = awareness of internal bodily states (hunger, pain, heartbeat).

Evolutionary hypothesis: Consciousness may have evolved to monitor the body for survival signals. Later, these internal feelings became directed outward as social emotions (disgust, shame, etc.).

3.2.8 Information-Theoretic Theories of Consciousness

Many features of consciousness can be described in **informational terms**:

- Meta-awareness = information about information
- Integration of information from multiple sources
- Making information widely available

This suggests theories that characterize consciousness in terms of **information processing**—without necessarily solving the Hard Problem.

Definition(Global Workspace Theory (GWT)).

Core idea: Consciousness arises from a **competition** among processors. "Winning" outputs are **broadcast** to a global workspace, making them widely accessible.

Mechanism:

1. Multiple processors compete for attention.
2. Winning outputs are broadcast to the global workspace.
3. Other subsystems receive and respond to the broadcast.

4. Feedback from subsystems **strengthens** the broadcast content.

Result: The broadcast information becomes **access-conscious**—available for use by diverse mental subsystems.

Definition(Integrated Information Theory (IIT) — Tononi).

Core idea: Consciousness is linked to the degree of **information integration**, measured by Φ (phi).

Three criteria for consciousness:

1. **Information about itself:**
   - The information that matters is what the system has about **its own causal powers**.
   - How much can we know about the system's past/future state from its current state?
   - Brain: high (current state tells us a lot about past/future states).
   - Retina: low (state determined by external input, not its own history).
2. **Integration:**
   - Φ measures how much information is **lost** by cutting the system in half.
   - Book page: tear in half $\rightarrow$ almost no information lost (not integrated).
   - Brain: cut in half $\rightarrow$ massive information loss (highly integrated).
3. **Maximality:**
   - A conscious system must have **more** integrated information than any of its parts or any larger system it belongs to.

Key Argument(IIT's Key Claims).

- High $\Phi \rightarrow$ high consciousness.
- In principle, we could build a "consciousness meter" that measures Φ.
- **Panpsychism:** IIT implies that even simple systems have tiny amounts of proto-consciousness (extremely low Φ).
- **Computers have low** Φ because transistors are sparsely connected, feed-forward, not recurrent. Therefore, IIT **rejects functionalism**—same I/O doesn't guarantee consciousness.

Example(Shanahan's Challenge to IIT).

Imagine gradually replacing neurons with **functionally equivalent digital components**, simulating all connections.

Would consciousness gradually disappear? IIT predicts yes (low integration). But this seems counterintuitive—the system still functions identically.

Alternative: Future computers might be built with sufficient integration (recurrent, densely connected architectures).

3.2.9 Could AI Be Conscious?

Key Argument(Criteria for Machine Consciousness).

How would we recognise that an AI is conscious?

1. **Functionalist criterion:** Acts/behaves as if conscious.
2. **Information-theoretic measures:** High Φ or other integration metrics.
3. **Architecture:** Right kind of structure (recurrent, densely interconnected).

Caveat: We can *never* definitively prove a machine is conscious—we can't experience its inner life. But we might have strong circumstantial evidence.

Definition(Substrate Independence of Consciousness).

If consciousness = complex information processing (integration, availability, etc.), and information processing is **substrate-neutral**, then:

Consciousness could be substrate-independent.

AI built with the right architecture might be conscious—regardless of being silicon-based rather than carbon-based.

Exam Tip(Open Question).

Would we design consciousness, or would it emerge?

- **Designed:** Intentionally build AI with high Φ, right architecture.
- **Emergent:** Consciousness naturally arises as AI becomes more sophisticated and processes information in increasingly complex ways.

3.2.10 Summary: Key Takeaways

Key Argument(Week 3 Key Points).

1. **Consciousness** = subjective experience; "what it is like" to be something.
2. **Easy Problem:** How the brain works (mechanism). **Hard Problem:** How physical matter gives rise to experience.
3. **Dualism:** Conscious properties are not physical. **Physicalism:** Everything is physical; consciousness is explainable.
4. **Functionalism:** Mental states defined by functional role, not substrate.
5. **Zombie argument:** If zombies are conceivable/possible, physicalism is false.
6. **Qualia:** Raw experiential properties. **Access consciousness:** Information available to other mental subsystems.
7. **GWT:** Consciousness = broadcast to global workspace.
8. **IIT:** Consciousness = integrated information (Φ). Implies panpsychism; rejects functionalism for current computers.
9. If consciousness is substrate-independent, **AI could be conscious**.

Definition(Key Terms to Remember).

- **Consciousness:** Subjective experience; "what it is like."
- **Easy/Hard Problem:** Mechanism vs. experience.
- **Qualia:** Raw sensory feels.
- **Access consciousness:** Information availability.
- **Dualism / Physicalism / Functionalism:** Theories of mind.
- **Philosophical Zombie:** Physical duplicate without experience.
- **Intentionality:** "Aboutness"—mental states relating to the world.
- **Epiphenomenalism:** Consciousness has no causal role.
- **Interoception:** Awareness of internal bodily states.
- **GWT:** Global Workspace Theory.
- **IIT:** Integrated Information Theory.
- Φ **(Phi):** Measure of information integration.
- **Panpsychism:** All matter has some (proto-)consciousness.

4 Week 4: Reasoning and Communication

4.1 Topic Summary

Key Points(Learning Objectives).

By the end of this week, you will be able to:

1. Explain **logic-based (symbolic) AI** and its foundations.
2. Describe **machine learning** approaches and their successes.
3. Identify **limitations** of both paradigms.
4. Understand **non-monotonic reasoning** and why it matters.
5. Explain **argumentation theory** as a formalization of reasoning.
6. Describe how **dialogue** enables distributed reasoning among multiple agents.

The Big Picture

AI has two major paradigms: **logic-based (symbolic)** and **machine learning**. Each has strengths and limitations. Logic-based AI struggles with real-world complexity and brittleness; ML struggles with explainability, causation, and common-sense reasoning. **Argumentation theory** offers a middle ground—formalizing non-monotonic reasoning in a way that is understandable to humans, computationally tractable, and supports **joint reasoning via dialogue**. This has important ethical implications: transparent, collaborative AI-human reasoning.

What Examiners Like to Ask

- Compare logic-based AI and machine learning approaches.
- What is monotonic vs. non-monotonic reasoning? Why does it matter?
- Explain the symbol grounding problem.
- What are the limitations of machine learning (black box, no causation, etc.)?
- What is argumentation? How do arguments attack each other?
- Explain Dung's argumentation framework and the labeling approach.
- What are grounded and preferred extensions?
- How can argumentation support distributed reasoning via dialogue?

4.2 Overview + Core Concepts + Extended Theory

Roadmap

1. Logic-based (symbolic) AI
2. Machine learning AI
3. Limitations of both paradigms

4. Non-monotonic reasoning
5. Argumentation theory (Dung's framework)
6. Distributed reasoning via dialogue

4.2.1 Logic-Based (Symbolic) AI

Definition(Good Old-Fashioned AI (GOFAI)).

Early AI used **symbolic formulae** to represent beliefs, desires, and goals. Reasoning involved applying **inference rules** to derive new facts about the world and determine appropriate actions.

Key idea: Describe the world in a logical language (like natural language describes states of affairs), then apply inference rules to deduce new truths and actions.

Logic in Mathematics (Background)

- Start with **self-evidently true axioms** (e.g., $\alpha \vee \neg\alpha$).
- Apply **inference rules** (e.g., disjunctive syllogism: from $\alpha \vee \beta$ and $\neg\alpha$, infer β).
- Derive complex truths from simple axioms.
- **Example:** Euclid proved all truths of 2D geometry from 5 basic axioms.

Definition(Soundness and Completeness).

- **Sound:** Everything inferred from axioms is true in the model.
- **Complete:** Every true statement about the model can be inferred.

Ideally, a logical system is both sound and complete.

Logic in AI

- Designer encodes **axiomatic truths** Δ_0 about the world in the agent's **knowledge base** (KB).
- Agent uses a **proof system** (PS) to infer new facts.
- Given goals, agent **acts**, bringing about changes to the world.
- Agent **senses** new facts Δ_1, updates KB, and the cycle continues.

Key Argument(Reductionist Arguments for Logic).

1. A complete proof system for first-order logic can simulate all Turing-level computation $\rightarrow$ achievement of any complex goal.
2. If intelligence can be formalized mathematically, and mathematics has logical foundations, then intelligence can be formalized in logic.

Modal Logics and BDI **Definition(Modal Logic).**

Extends classical logic with **modalities** that qualify statements:

- **Alethic modalities:** $\Box p$ = necessarily p (true in all possible worlds); $\Diamond p$ = possibly p (true in at least one possible world).
- **Deontic modalities:** $Oq = q$ is obligatory; $Pq = q$ is permitted; $Fq = q$ is forbidden.

Deontic logic is relevant to AI ethics (reasoning about what agents *ought* to do).

Definition(BDI Logic (Belief-Desire-Intention)).

Based on Bratman's model of practical reasoning:

- **Beliefs:** What the agent believes about the world (may not be true).
- **Desires:** Motivational states (what agent would like to accomplish); can be inconsistent.
- **Goals:** Desires being actively pursued; must be consistent.
- **Intentions:** High-level plans to achieve chosen goals.

Example inference: $\text{Intend}(q) \to \text{Bel}(\neg q)$ — if you intend to bring about q, you must believe q is not already the case.

4.2.2 Machine Learning AI

Definition(Machine Learning (Definition)).

"A computer program is said to **learn from experience E** with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, **improves with experience E**."

Type	Description
Supervised	Learns from labeled data (inputs + desired outputs).
Semi-supervised	Learns from partially labeled data.
Unsupervised	Learns structure from unlabeled data (clustering, etc.).
Reinforcement	Learns from feedback (rewards/punishments) in a dynamic environment.

Key Argument(Deep Learning).

Deep learning uses multi-layer neural networks where each layer transforms input into increasingly **abstract representations**. Responsible for major AI successes:

- AlphaGo, AlphaZero (game playing)
- Translation, speech recognition
- Computer vision, medical imaging
- Text generation (GPT-3, etc.)

4.2.3 Limitations of Both Paradigms

Issue	Logic-Based AI	Machine Learning
Scalability	Too many rules needed; doesn't scale from toy examples to real world.	Needs very large datasets.
Brittleness	Small damage → complete failure.	More robust; degrades gracefully.
Sensory tasks	Not designed for low-level perception.	Excels at perception tasks.
Symbol grounding	Hand-crafted representations; not grounded in real-world data.	Learns from data; solves grounding.
Explainability	Transparent (symbolic).	**Black box**; opaque reasoning.
Hierarchical structure	Natural (recursive rules).	Struggles with hierarchy.
Causal reasoning	Can represent causation.	Learns correlation, not causation.
Common sense	Theoretically addressed by non-monotonic logic, but computationally hard.	Struggles with common-sense inference.
Generalization	Explicit rules generalize.	**Transfer** problems; poor on novel tasks.
Joint reasoning	Not designed for multiple agents.	Hard to integrate with human reasoning.

Definition(The Symbol Grounding Problem).

In logic-based AI, symbols are **hand-crafted by designers** rather than grounded in real-world data.

Problem: The semantics (meaning) of symbols relies on the designer's mind, not on direct connection to the world. This prevents ongoing adaptation to unknown environments and is a barrier to full autonomy.

ML solution: Machine learning "grounds" representations by extracting them from large datasets.

Definition(The Black Box Problem).

ML reasoning processes are **opaque**—even designers can't explain how the system reaches its conclusions.

Consequences:

- Cannot provide symbolic/language-like explanations.
- Cannot integrate human and machine reasoning.
- Has important **ethical implications** (accountability, trust, bias detection).

4.2.4 Non-Monotonic Reasoning

Definition(Monotonic vs. Non-Monotonic Logic).

Classical logic is monotonic:

$$\mathrm{KB} \cup \Delta_0 \vdash_{\mathrm{PS}} \alpha \implies \mathrm{KB} \cup \Delta_0 \cup \Delta_1 \vdash_{\mathrm{PS}} \alpha$$

Adding new information never invalidates previous conclusions.

Problem: The real world is too complex to encode all exceptions. We use **rules of thumb** that can be overridden.

Non-monotonic logic: Conclusions can be **withdrawn** when new information conflicts with them.

Example(Tweety the Penguin).

- KB contains: "Birds typically fly"; "Tweety is a bird."
- Infer: Tweety flies.
- New information: "Tweety is a penguin."
- Withdraw previous conclusion; infer: Tweety does **not** fly.

The inference ¬fly(Tweety) is **preferred** over fly(Tweety) because penguins are a **subclass** of birds (specificity principle).

Definition(Defeasible Rules).

Non-monotonic logics use **defeasible rules**:

- "Birds *typically/usually* fly" (rather than "necessarily").
- "Birds fly *unless there is evidence to the contrary.*"

These rules can be overridden by more specific information.

Exam Tip(Relevance to AI Ethics).

The problem of exceptions is crucial for ethical AI. You can specify a moral rule (e.g., "Do not harm humans"), but there will always be exceptions. How do you encode all possible exceptions? Non-monotonic reasoning and argumentation address this.

4.2.5 Argumentation Theory

Definition(Why Argumentation?).

Non-monotonic reasoning = deciding among **conflicting inferences** (beliefs, goals, actions).

Argumentation formalizes this as:

- More in line with how humans reason (debate, discussion).
- Can be formalized to give rational outcomes under limited resources.
- Enables **joint reasoning** among multiple agents (human and AI).
- Reasoning becomes **publicly visible**, not hidden in a black box.

Definition(Arguments and Attacks).

An **argument** is a pair: $(\text{Premises}, \text{Claim})$.

Example:

- $A_1 = (\{\text{penguin}(\text{Tweety}), \text{penguin} \to \neg\text{fly}\}, \neg\text{fly}(\text{Tweety}))$
- $A_2 = (\{\text{bird}(\text{Tweety}), \text{bird} \to \text{fly}\}, \text{fly}(\text{Tweety}))$

A_1 and A_2 **attack** each other (contradictory claims).

If A_1 is **preferred** (specificity: penguins are a subclass of birds), then A_2's attack on A_1 is cancelled. A_1 wins.

Dung's Argumentation Framework (1995) **Definition(Argumentation Framework).**

An **argumentation framework** is a pair $\langle \text{Args}, \text{Attacks} \rangle$ where:

- Args = set of arguments.
- Attacks $\subseteq$ Args $\times$ Args = binary attack relation.

Represented as a **directed graph**: nodes = arguments, edges = attacks.

Definition(Labeling Arguments).

Label each argument as:

- **IN** (winning): All attackers are OUT.
- **OUT** (losing): At least one attacker is IN.
- **UNDEC** (undecided): At least one attacker is UNDEC, and no attacker is IN.

A **legal labeling** satisfies these conditions for all arguments.

Example(Labeling Example).

Framework: $A_3 \to A_2 \to A_1$ (i.e., A_3 attacks A_2; A_2 attacks A_1).

Legal labeling:

- A_3: IN (no attackers).
- A_2: OUT (attacker A_3 is IN).
- A_1: IN (all attackers—just A_2—are OUT).

Definition(Grounded vs. Preferred Extensions).

Given all legal labelings:

Grounded extension:

- The unique labeling with **minimum** IN arguments.
- Most skeptical/cautious.

Preferred extensions:

- Labelings with **maximal** IN arguments (no superset exists).
- May be multiple preferred extensions.
- More credulous.

Example(Mutual Attack: $A_1 \leftrightarrow A_2$).

Three legal labelings:

1. A_1: IN, A_2: OUT — *preferred*.
2. A_1: OUT, A_2: IN — *preferred*.
3. A_1: UNDEC, A_2: UNDEC — *grounded* (minimum IN = $\emptyset$).

Interpretation: A genuine dilemma. Preferred extensions choose one side; grounded extension remains skeptical.

4.2.6 Argument Games and Dialogue

Definition(Argument Game Proof Theory).

To determine if argument X is in the grounded/preferred extension:

1. **PRO** (proponent) moves X.
2. **OPP** (opponent) moves an attacking argument.
3. PRO and OPP alternate, moving attacks according to game rules.
4. If PRO successfully counters all OPP moves, X is **justified** (in the extension).

Rules differ:

- **Grounded game:** PRO cannot repeat arguments in a dispute.
- **Preferred game:** OPP cannot repeat arguments in a dispute.

Definition(From Single-Agent to Distributed Reasoning).

Single-agent: Agent constructs framework from private KB, uses argument game to find winning arguments.

Multi-agent (dialogue): Multiple agents exchange arguments publicly. The framework is built from **publicly communicated contents**, not private KBs.

Public semantics: Argument X wins iff X is in an extension of the framework built from what has been **publicly communicated**.

Key Argument(Dialogical Reasoning).

A dialogue that establishes α effectively demonstrates that α can be inferred from the contents of the arguments exchanged during the dialogue.

Implications:

- Reasoning is **transparent** (publicly visible).
- Multiple minds bring different knowledge and perspectives.
- Enables **human-AI joint reasoning**.
- Important for ensuring **ethical AI decisions**.

4.2.7 Summary: Key Takeaways

Key Argument(Week 4 Key Points).

1. **Logic-based AI:** Symbolic reasoning; sound/complete proof systems; BDI logics for beliefs, desires, intentions.
2. **Machine learning:** Learns from data; deep learning enables abstraction; major successes in perception/generation.
3. **Limitations:**
   - Logic: Brittleness, symbol grounding, scalability.
   - ML: Black box, no causation, poor transfer, no joint reasoning.
4. **Non-monotonic reasoning:** Conclusions can be withdrawn; uses defeasible rules; addresses real-world exceptions.
5. **Argumentation:** Arguments attack each other; preferences determine winners; labelings (IN/OUT/UNDEC).
6. **Grounded extension:** Minimum IN (skeptical). **Preferred extensions:** Maximal IN (credulous).
7. **Dialogue:** Enables distributed reasoning; framework built from public communication; transparent, joint AI-human reasoning.
8. **Ethical relevance:** Argumentation supports explainable, collaborative reasoning—crucial for ethical AI.

Definition(Key Terms to Remember).

- **GOFAI:** Good Old-Fashioned AI (symbolic/logic-based).
- **Soundness/Completeness:** Proof system properties.
- **Modal logic:** Modalities ($\Box$, $\Diamond$, O, P, F).
- **BDI:** Belief-Desire-Intention.
- **Symbol grounding problem:** Symbols not connected to world.
- **Black box problem:** Opaque ML reasoning.
- **Monotonic/Non-monotonic:** Whether new info can retract conclusions.
- **Defeasible rules:** Rules that can be overridden.
- **Argumentation framework:** $\langle \text{Args}, \text{Attacks} \rangle$.
- **IN/OUT/UNDEC:** Argument labels.
- **Grounded extension:** Minimal IN (skeptical).
- **Preferred extension:** Maximal IN (credulous).
- **Dialogical reasoning:** Joint reasoning via public argument exchange.

5 Week 5: Ethics and Morality

5.1 Topic Summary

Key Points(Learning Objectives).

By the end of this week, you will be able to:

- Identify key **ethical challenges** posed by AI.
- Explain the main **philosophical theories of ethics**: deontology, consequentialism, virtue ethics.
- Describe **Moor's categorization** of moral agents.
- Compare **top-down** (deontic logic) and **bottom-up** (ML) approaches to implementing moral agents.
- Analyze **utilitarianism** and its critiques using trolley problem thought experiments.
- Discuss how ethical theories might be **combined** in AI implementations.

The Big Picture

Ethics concerns what we *ought* to do. As AI becomes more autonomous, we must ensure AI systems behave ethically. This week covers three major ethical theories—deontology (rules/duties), consequentialism (outcomes), and virtue ethics (character)—and how they inform the design of **artificial moral agents** (AMAs). Utilitarianism (a form of consequentialism) is explored in depth, including how human moral intuitions can diverge from rational utilitarian calculations.

What Examiners Like to Ask

- Compare and contrast deontology, consequentialism, and virtue ethics.
- What is Kant's Categorical Imperative? What are its critiques?
- Explain Moor's categorization of moral agents (ethical impact, implicit, explicit, full).
- How do top-down and bottom-up approaches to AMAs differ?
- What is utilitarianism? How is it defined?
- Analyze the trolley problem variations: What do they reveal about moral intuitions?
- Distinguish between "means to an end" and "side effect" in moral reasoning.
- What are the critiques of utilitarianism? How might they be addressed?
- How might ethical theories be combined in AI implementations?

5.2 Overview + Core Concepts + Extended Theory

Roadmap

1. Ethical challenges for AI
2. Key theories of ethics: Deontology, Consequentialism, Virtue Ethics

3. Moral relativism
4. Moor's categorization of moral agents
5. Implementing artificial moral agents (top-down vs. bottom-up)
6. Utilitarianism in depth
7. Trolleyology: Moral intuitions vs. utilitarian calculation
8. Combining theories for AMAs

5.2.1 Ethical Challenges for Artificial Intelligence

Key Points(Why Are We More Worried Now?).

1. **Non-segregated environments:** AI (e.g., autonomous cars) now operates in open, unpredictable spaces, not just restricted areas with trained personnel.
2. **Increased autonomy:** AI makes decisions independently, without human intervention.
3. **Unpredictable contexts:** ML enables deployment in situations designers cannot fully anticipate.
4. **Lack of transparency:** ML algorithms are "black boxes"—we can't explain how they decide.

Example(Examples of Ethical Challenges).

- **Autonomous cars:** Brake failure → injure 3 workers or swerve and injure driver?
- **Autonomous weapons:** R&D in US, UK, China, Russia, etc.
- **ML in decisions:** Insurance, recruitment, mortgages, criminal justice → bias, fairness, transparency.
- **Social robots:** Sex robots, elderly care robots → anthropomorphization, ethical treatment.
- **Social media / IoT:** Data privacy, surveillance, manipulation.
- **Filtering algorithms:** Polarizing beliefs, targeted advertising, restricting choices.
- **Medical AI:** Decision support, surgical robots, diagnosis.
- **Superintelligence:** Value loading problem (AI goals misaligned with human values).

Definition(Ethics and AI: Key Questions).

1. How should **we** use AI ethically? (User responsibility)
2. How do we **design** AI so it cannot act unethically? (Hardcoded constraints)
3. How do we design **autonomous** AI to *choose* ethically in unforeseen situations?
4. How do we remain **aware** of AI's ethical impact as it becomes embedded in society?

5.2.2 Key Theories of Ethics and Morality

Theory	Core Idea
Deontology	Focus on **duties and rights**; morality = following rules/norms/laws.
Consequentialism	Focus on **outcomes**; choose actions with the best consequences.
Virtue Ethics	Focus on **character**; cultivate virtues (honesty, courage, etc.) → act ethically.

Example(Helping Someone in Need).

- **Consequentialist:** Helping maximizes well-being.
- **Deontologist:** Helping accords with "Treat others as you would want to be treated."
- **Virtue Ethicist:** Helping is what a charitable/benevolent person would do.

Deontology **Definition(Deontology).**

Concerned with choices that are morally **forbidden** (prohibited), **required** (obligated), or **permitted**. Guides what we *ought* to do by following rules that encode duties and respect rights.

Two approaches:

- Agent-focused: Duties of the agent to be moral.
- Rights-focused: Respect for rights of others.

Example(Deontological Rules).

Specific rules:

- Religious: Ten Commandments, Hindu prohibition on beef.
- Secular: Laws against murder, theft; Asimov's Laws of Robotics.

General principles:

- **The Golden Rule:** Treat others as you would like to be treated.
- **Kant's Categorical Imperative (CI):** Act only according to that rule by which you can at the same time will that it should become a *universal law*.

Definition(Kant's Categorical Imperative).

Formulation 1: Act only according to that rule by which you can *will* that it should become a **universal law**.

Example: If you consider lying to get what you want, you must will that *everyone* lies. But then no one believes anyone, and lying fails. So lying is not permitted—it requires making an exception for yourself.

Formulation 2: Act so that you treat humanity (yourself and others) always as an **end** and never simply as a **means**.

→ Slavery is wrong because slaves are used as means to the slave owner's ends.

Exam Tip(Critiques of Deontology).

1. **Too general/vague:** Doesn't always guide action in specific situations.
2. **Inflexible:** Absolute prohibitions; what about exceptions? (Can update rules, but then need to list all exceptions—computationally infeasible.)
3. **No priority among rules:** When duties/rights conflict, no clear resolution. (E.g., abortion: right to life vs. right to choose.)
4. **Harmful consequences:** Following rules may lead to bad outcomes. (E.g., "A sadist is a masochist who follows the Golden Rule.")

Virtue Ethics **Definition(Virtue Ethics).**

Morally good actions flow from the cultivation of **good character** (virtues): honesty, courage, compassion, wisdom, justice, etc.

Origins: Socrates, Aristotle.

Learning: What is good is learned through intuition, induction, experience—observing good people, being rewarded for good behavior.

→ Resonates with **machine learning**: learn ethical behavior through experience and reinforcement, not explicit rules.

Exam Tip(Critiques of Virtue Ethics).

- Different cultures have different virtues; no objectively "right" list.
- Does not provide guidance on which actions are permitted/prohibited, only on what character to cultivate.
- In novel situations, virtues may not give definitive guidance (same problem for ML agents).

Consequentialism **Definition(Consequentialism).**

Actions should be judged **only by their consequences**. Specify what is "good," then choose actions that maximize the good.

No rules or rights: All that matters is outcomes. (E.g., lying to save an innocent life may be justified.)

Fits AI/AIXI model: Agent maximizes expected utility (= the good).

5.2.3 Moral Relativism

Definition(Moral Relativism).

Argument:

1. Morality = value judgments in a given society.
2. Other societies have their own judgments; we have ours.
3. **Conclusion:** We should not judge other cultures.

Critique: The conclusion is itself a value judgment announced as a universal truth ("no matter what culture, you should not judge"). This is self-contradictory.

5.2.4 Moor's Categorization of Moral Agents

Category	Description
Ethical Impact Agent	Any machine whose operations increase or decrease good in the world. (E.g., robot camel jockeys replacing child jockeys; search ranking algorithms.)
Implicit Ethical Agent	Designed not to have negative ethical effects; behavior constrained by designers. No explicit representation of ethics. (E.g., pilot collision warning systems; robotic lawnmowers that can't harm.)
Explicit Ethical Agent	Can reason about the best action in ethical dilemmas, including **novel** situations. Uses explicit representations of ethics (e.g., deontic logic rules).
Full Ethical Agent	Has **moral agency**: understands ethical impacts, freely chooses, can justify moral judgments. → Morally responsible and accountable.

Exam Tip(Key Question).

Are these categories mutually exclusive? (E.g., an implicit ethical agent may also be an ethical impact agent.)

5.2.5 Implementing Artificial Moral Agents (AMAs)

Approach	Description	Correspondence
Top-Down	Encode moral theory (rules) in agent; agent applies rules to act ethically. Logic-based, deontic.	Deontology
Bottom-Up	Agent learns ethical behavior through experience (reinforcement learning). Values implicit, not articulated.	Virtue Ethics
Hybrid	Combines top-down and bottom-up.	Multiple theories

Top-Down: Deontic Logic Implementations **Definition(Deontic Logic).**

Uses modalities:

- $Oq = q$ is **obligatory**.
- $Pq = q$ is **permitted**.
- $Fq = q$ is **forbidden**.

Example(Autonomous Car Example (UK)).

- $F(\text{drive_on_right})$ — forbidden to drive on right.
- $O(\text{protect_driver})$ — obligated to protect driver.

Scenario: In a tunnel, collision imminent; only way to avoid is to swerve right.

Conflict: Must prioritize rules.

Rule: If collision imminent and risk of injury on left > risk on right, then $O(\text{protect_driver}) > F(\text{drive_on_right})$.

Problem: What if not in a tunnel? What about risk to others? What if there's a child in oncoming car? → Must encode *all* exceptions—computationally infeasible.

Key Argument(The Core Problem).

For autonomous agents in complex, unpredictable environments:

- Deontic rules must be **general** (can't anticipate all situations).
- Challenge: How to **interpret** general principles in specific situations?
- Related to **commonsense reasoning** problem (Week 4).

Example(Asimov's Laws of Robotics).

0. A robot may not injure humanity, or, by inaction, allow humanity to come to harm.
1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.

2. A robot must obey orders given by human beings except where such orders conflict with the First Law.
3. A robot must protect its own existence as long as this does not conflict with the First or Second Laws.

Problem: What does "harm" mean? A literal-minded robot might interrupt a surgeon cutting a patient. Unless context is fully described, rules can lead to unintended consequences.

Bottom-Up: Machine Learning Implementations

- Agent learns ethical behavior through **reinforcement learning**.
- Equates with **virtue ethics**: learning through experience, not explicit rules.
- Agent maximizes rewards by optimizing utility functions.
- **Problem:** In novel situations, no past experience to draw on.

5.2.6 Utilitarianism

Definition(Utilitarianism).

Utilitarianism = Consequentialism + Happiness as the measure of good.

$$\text{Utilitarianism} = \text{Impartial Maximization of Happiness}$$

- **Impartial:** Everyone's happiness counts equally (related to Golden Rule).
- **Maximization:** Choose actions that produce the greatest net happiness.

Founders: Jeremy Bentham, John Stuart Mill (18th–19th century).

Key Points(Why Happiness?).

1. **Universal:** True for everyone → basis for universal morality.
2. **Foundational:** All values we cherish (family, friendship, freedom, justice) are ultimately linked to happiness. Subtract their positive impact on experience, and their value is lost.

What is happiness?

- Subjective experience of well-being.
- Counterfactual test: If something we value were absent, would happiness decrease? Then it contributes to happiness.

Key Argument(Act vs. Rule Utilitarianism).

- **Act Utilitarianism:** Apply utility calculation case-by-case; choose action that maximizes utility in that situation.
- **Rule Utilitarianism:** Actions are justified if they conform to *rules* that, if included in our moral code, would maximize utility overall. (Combines deontology and utilitarianism.)

5.2.7 Trolleyology: Moral Intuitions vs. Utilitarian Calculation

Definition(The Trolley Problems).

Thought experiments involving a trolley hurtling toward workers tied to a track. You know nothing about the workers (age, character, etc.). The question: What would you do?

The Switch Dilemma (Standard) A trolley is heading toward 5 workers. At point A, you can steer the trolley onto a side track, killing 1 worker instead of 5.

Response: 87% would pull the switch (kill 1, save 5).

Key features:

- No physical force (just pull a switch).
- Death of 1 is a **side effect**, not a means to an end. (Counterfactually: if the 1 worker weren't there, you'd still save the 5.)

The Footbridge Dilemma You are on a footbridge. A large man with a backpack is next to you. The only way to stop the trolley is to push him off the bridge; he dies but stops the trolley, saving 5.

Response: Only **31%** would push the man.

Why the difference? (Same utilitarian calculation: kill 1, save 5.)

1. **Physical force:** Actively pushing someone → emotional resistance to violence (evolved to preserve social cooperation).
2. **Means to an end:** The man is used as a "trolley stopper." His death is *causally necessary* to save the 5. (Counterfactually: if you didn't push him, you wouldn't save the 5.)

Footbridge + Switch Dilemma You are on a footbridge. You can run to the end and pull a switch to divert the trolley, saving 5. But you must collide with the man, knocking him off the bridge (he dies).

Response: 81% would run to pull the switch.

Why?

- Physical force: Yes.
- Means to an end: **No**—the man's death is a **side effect**. (Counterfactually: if he weren't there, you'd still save the 5.)

Switch Dilemma (Variant) Same as standard, but this time, if the 1 worker were not on the side track, the trolley would loop back and kill the 5. So the 1 worker is used as a trolley stopper (means to an end).

Response: 81% would steer right.

Why?

- No physical force.
- Means to an end: Yes, but without physical force, the aversion is reduced.

Key Argument(Key Insight: Physical Force + Means to End).

It is the **combination** of:

1. **Physical force** (violence), and
2. **Means to an end** (death is causally necessary for the goal)

that triggers strong emotional resistance and interferes with utilitarian calculation.

Why?

- **Physical force:** We evolved to avoid violence (disrupts cooperation, threatens survival).
- **Means to an end:** When planning actions, we cognitively "see" the causal chain. Harms that are *means* are more salient → trigger emotional alarm. *Side effects* are less salient.

Example(International Law: Means vs. Side Effect).

- **Illegal:** Intentionally bombing civilians to lower enemy morale (civilians = means to an end).
- **Legal:** Bombing a munitions factory, knowing civilians will die as "collateral damage" (side effect).

Critique: Both are foreseeable deaths. Why is there a moral distinction? The distinction between means and side effect is arguably **morally irrelevant**—but our intuitions are shaped by this cognitive bias.

Key Argument(Trolleyology Conclusion).

- People's moral intuitions diverge from pure utilitarian calculation.
- This is due to evolved emotional responses (aversion to violence, salience of means vs. side effects).
- But what people *emotionally think* is right doesn't make it right.
- **AIs** will not have these evolutionarily acquired biases. They could apply utilitarian calculations without emotional interference.

5.2.8 Critiques of Utilitarianism

Critique	Example	Response
Not demanding enough (leads to unethical choices)	5 people need organ transplants. Kill 1 healthy person to harvest organs and save 5?	Which society is happier: one where this is legal, or one where it is not? The latter → utilitarianism doesn't actually endorse this.
Too demanding (requires unreasonable sacrifices)	Instead of buying a $500 suit, donate to save children in poverty.	**Pragmatic utilitarianism:** Being a "perfect" utilitarian is anti-utilitarian (incompatible with human nature). Be a role model others can emulate → greater good.

Exam Tip(Utilitarianism and the Abortion Debate).

- Pro-choice: Appeals to **right to choose**.
- Pro-life: Appeals to **right to life**.

Both are deontic rights → no clear resolution when they conflict.

Utilitarian approach: Appeal to **evidence** about consequences. Does legalizing abortion increase or decrease overall well-being? (E.g., reduced poverty, fewer backstreet abortions, reduced crime?) This can be studied empirically.

When moral issues are **settled** (e.g., slavery), express them as rights. But for **controversial** dilemmas, utilitarianism may offer a path forward.

5.2.9 Combining Ethical Theories for AMAs

Key Argument(Hybrid Approach).

Perhaps AMAs should combine all three ethical theories:

1. **Machine Learning (Virtue Ethics):** Reward agents for ethically good behaviors. Equip them to do the right thing in **familiar** situations with little controversy.
2. **Utilitarianism (Consequentialism):** For **novel, controversial** dilemmas, use utilitarian calculations (under bounded resources). AI can:
   - Collect and analyze data/evidence.
   - Model causal reasoning (superior to humans).
   - Humans provide input on how actions impact well-being (subjective experience).
3. **Rules (Deontology):** Once controversies are settled, encode as **rules** for long-term planning. (= Rule Utilitarianism.)

Exam Tip(Utilitarian AI: Computational Challenges).

- **Causal reasoning:** Computationally intractable to model all effects of actions.

- **Happiness estimation:** Must estimate effects on all conscious beings.
- **Bounded rationality:** Aim for "good enough" calculations under limited resources (like weather forecasting—imperfect but improving).

5.2.10 Summary: Key Takeaways

Key Points(Week 5 Key Points).

1. **Ethical challenges for AI:** Autonomy, unpredictability, lack of transparency.
2. **Deontology:** Duties, rights, rules (Golden Rule, Kant's CI). Critiques: inflexible, can't handle exceptions, no priority among rules.
3. **Virtue Ethics:** Cultivate good character. Critiques: no guidance on specific actions, culture-dependent.
4. **Consequentialism:** Judge actions by outcomes. Utilitarianism = impartial maximization of happiness.
5. **Moral relativism:** Self-contradictory.
6. **Moor's categories:** Ethical impact, implicit, explicit, full ethical agents.
7. **Top-down AMAs:** Deontic logic; problems with commonsense reasoning and exceptions.
8. **Bottom-up AMAs:** ML/reinforcement learning; problems in novel situations.
9. **Trolleyology:** Physical force + means-to-end → interferes with utilitarian calculation. Our intuitions are biased; AI could reason without these biases.
10. **Hybrid AMAs:** Combine ML (familiar situations), utilitarianism (novel dilemmas), rules (settled issues).

Definition(Key Terms).

- **Deontology:** Ethics based on duties/rules.
- **Categorical Imperative:** Act only on rules you can will as universal laws.
- **Virtue Ethics:** Ethics based on character.
- **Consequentialism:** Ethics based on outcomes.
- **Utilitarianism:** Impartial maximization of happiness.
- **Act vs. Rule Utilitarianism:** Case-by-case vs. rule-based utility.
- **Means to an end vs. Side effect:** Causally necessary vs. incidental harm.
- **Implicit/Explicit/Full Ethical Agent:** Moor's categories.
- **Top-down/Bottom-up:** Rule-based vs. learning-based AMAs.

6 Week 6: Algorithms

6.1 Topic Summary

Key Points(Learning Objectives).

By the end of this week, you will be able to:

- Explain the importance of **accountability and transparency** in algorithmic decision-making.
- Describe **procedural regularity** and techniques for ensuring it.
- Summarize the **ACM Principles** for algorithmic transparency and accountability.
- Analyze **liability and responsibility** frameworks for AI systems.
- Discuss **moral responsibility**, free will, and their implications for AI.
- Define **algorithmic bias** and identify its types (data, algorithm, use, feedback).
- Evaluate **solutions** to algorithmic bias (statistical, discursive, documentation, standards).

The Big Picture

As algorithms increasingly make decisions that affect lives (loans, sentencing, hiring, policing), ensuring **accountability** becomes critical. This week examines how we assign responsibility when algorithms cause harm, the philosophical issues of moral responsibility and free will (and why they may not apply to AI), and the pervasive problem of **algorithmic bias**—how societal biases become encoded in data and perpetuated by ML systems.

What Examiners Like to Ask

- Why is transparency important for accountability? When might transparency be undesirable?
- What is procedural regularity? What techniques ensure it?
- List and explain the ACM Principles for algorithmic accountability.
- What is data provenance and why is it important?
- Explain the three scenarios for criminal liability (perpetrator via another, natural probable consequence, direct liability).
- How does moral responsibility relate to the notion of self and free will?
- Why might a consequentialist approach be appropriate for AI penalties?
- Define algorithmic bias. What are the main types?
- Give examples of algorithmic bias (COMPAS, PredPol, facial recognition).
- What solutions exist for mitigating algorithmic bias?

Exam Tip(Note on Examinable Content).

The section on "Human Selves and Free Will" through "Free Will is also an Illusion" is **not examinable**. However, the conclusions drawn (no distinct self/free will → consequentialist approach to penalties) are relevant.

6.2 Overview + Core Concepts + Extended Theory

Roadmap

1. Accountability and Transparency
2. Procedural Regularity
3. ACM Principles for Algorithmic Transparency and Accountability
4. AI, Liability, and Responsibility
5. Moral Responsibility, Free Will, and Justice
6. Algorithmic Bias: Definition and Types
7. Recent Examples of Bias
8. Solutions to Algorithmic Bias

6.2.1 Accountability and Transparency

Definition(Accountability).

The ability to demonstrate to the public, regulatory bodies, and courts that automated decisions comply with standards of **legal fairness** and **ethical fairness**. Requires assigning responsibility/blame when things go wrong.

Key Points(Why Accountability Matters).

- Many critical decisions are now made by algorithms: votes, loans, credit cards, police targeting, tax audits, immigration visas, sentencing.
- Tools designed to oversee *human* decision-makers often fail for computers (e.g., how do you judge the *intent* of software?).
- Automated systems can return incorrect, unjustified, or unfair results—especially critical for life/death decisions.

Definition(Transparency).

The ability to **examine and explain** how algorithms make decisions. Essential for:

- Demonstrating ethical fairness.
- Providing incentives for compliance with ethical standards.

Plato's Ring of Gyges: A ring that makes the wearer invisible → wearer commits wrongdoing (no accountability). Modern analogue: opaque AI systems used by powerful corporations. Transparency lessens this risk.

Exam Tip(Limits of Transparency).

- **Black box problem:** Explaining how ML algorithms decide is a major unsolved issue.
- **Privacy:** Explaining a decision may reveal private data (violates GDPR).
- **Gaming:** Transparency may be abused (e.g., tax cheats, terrorists learning to evade screening).

→ Demanding transparency requires **weighing harms and benefits**—consequentialist reasoning, not a universal rule.

6.2.2 Procedural Regularity

Definition(Procedural Regularity).

For AI in narrow, predictable contexts: each participant knows the **same procedure** was applied, and the procedure was not designed to disadvantage them specifically.

Requirements:

1. Same policy/rule used for each decision.
2. Decision policy fully specified *before* decision subjects were known.
3. Each decision is **reproducible** from the policy and inputs.
4. Any random inputs are beyond the control of interested parties.

Key Points(Techniques for Procedural Regularity).

- **Software Verification:** Logic-based techniques to prove a program satisfies **invariants** (facts always true regardless of state/input).
  - **Model Checking:** Exhaustively test all possible inputs to ensure invariant is never violated. (E.g., traffic light algorithm: never set both lights to green simultaneously.)
- **Cryptographic Commitments:** Ensure the same decision policy was used for each decision, especially when full transparency is undesirable. (Commit to a value while keeping it hidden; reveal later.)

6.2.3 ACM Principles for Algorithmic Transparency and Accountability

The Association for Computing Machinery (ACM) is the world's largest computing society. Their principles:

Principle	Description
Awareness	Stakeholders (owners, designers, users) should be aware of biases in design/implementation/use and potential harm.
Access & Redress	Regulators should enable questioning and redress for those adversely affected.
Accountability	Institutions are responsible for algorithmic decisions, even if the algorithm is a black box.
Explanation	Systems should produce explanations of procedures and specific decisions (especially in public policy).
Data Provenance	Document inputs, entities, systems, and processes that influence data; explore potential biases. Public scrutiny where possible (restrict access for privacy/security).
Auditability	Models, algorithms, data, and decisions should be recorded for auditing when harm is suspected.
Validation & Testing	Rigorous methods to validate models; routinely test for discriminatory harm; publish results.

Definition(Data Provenance).

A historical record of data's origins, what processing it underwent (with links to algorithms and parameters), and which systems/entities processed it.

Purpose: Enable exploration of biases; support public scrutiny; provide authoritative source for data integrity (combats "info-pocalypse" / fake news).

6.2.4 AI, Liability, and Responsibility

Key Points(The Problem).

With increasingly **autonomous** AI systems, who is responsible when things go wrong?

Example: Autonomous vehicles (AVs):

- Driver error causes 94% of conventional car crashes; AVs could improve this.
- But AVs still crash (30+ accidents in California since 2014).
- Current assumption: Companies behind software/hardware are liable, not car owner.
- But what if AV warned driver to disengage autopilot? Is human then responsible?
- Future AVs may have no steering wheel—carmakers must take full responsibility.

Criminal Liability Frameworks Criminal liability usually requires **action** and **mental intent**. Three scenarios:

Scenario	Description
Perpetrator via Another	Offense committed by mentally deficient person/animal (innocent agent). The *instructor* is criminally liable. (E.g., dog owner instructs dog to attack.) → AI as innocent agent; programmer/user as perpetrator.
Natural Probable Consequence	Programmer knew outcome was a probable consequence. (E.g., 1981: AI robot in Japanese factory killed worker by pushing him into machine—did programmer anticipate this?)
Direct Liability	Requires action + intent. Action is easy to prove; intent is harder. (Relates to moral responsibility.)

Civil Liability If criminal liability doesn't apply, the matter is settled in civil court. Key question: Is AI a **product** or a **service**?

	Product	Service
Legal Framework	Product design legislation (warranty).	Rules of negligence.
Plaintiff Must Prove	Warranty/guarantee violated.	(1) Defendant had duty of care; (2) Defendant breached duty; (3) Breach caused injury.

6.2.5 Moral Responsibility, Free Will, and Justice

Definition(Moral Responsibility).

A person is morally responsible for a crime if:

1. They were **conscious** (understood) that the crime would cause harm.
2. They were **free to do otherwise**—exercised **free will**.

Free will implies a **self** that freely chooses. This also characterizes Moor's **full ethical agent**.

Key Points(Functions of Penalties).

1. **Protection:** Protect society from criminal, and criminal from society (vigilante justice).
2. **Deterrence:** Deter others from committing the crime.
3. **Rehabilitation:** Ensure criminal doesn't re-offend.
4. **Retribution:** Punish the self for having freely chosen to commit the crime.

Note: (1)–(3) can be justified on **consequentialist** grounds (maximize good outcomes). (4) Retribution requires the assumption of a self that freely chose—and is *not* obviously consequentialist.

AI Selves and Free Will **Key Argument(No Self, No Free Will in AI).**

- AI is "just algorithms" processing information, maximizing utility, following rules.
- There is no distinct **self** ("ghost in the machine") that freely chooses.
- **Conclusion:** AI cannot be morally responsible in the traditional sense.

Exam Tip(Implications for AI Penalties).

Since AI has no self/free will:

- Use **consequentialist** framework to decide penalties.
- **Protection:** Take AI out of use.
- **Deterrence:** Penalize developers/companies to discourage similar designs.
- **Rehabilitation:** Modify the decision-making algorithm.
- **Retribution:** Doesn't make sense—no self to punish.

However, we may still *treat* AI as if morally responsible (to maintain public trust). If AI commits a serious crime, the public may only trust AI if it is visibly "punished" (decommissioned).

Human Selves and Free Will (Not Examinable, but Relevant) **Key Points(Summary (Non-Examinable))**.

The same reasoning applies to humans:

- Neuroscience, philosophy, and Buddhism suggest the **self is an illusion**—just neurons firing, no CEO/self weaving together experience.
- **Free will** is also illusory: choices are determined by prior causes (genes + environment + physics).
- The *feeling* of freely choosing is the conscious experience of a deterministic process.

Implication: Abandon retribution; focus on consequentialist penalties (protection, deterrence, rehabilitation). But treating people *as if* they have free will serves a useful social function.

Key Argument(Does "Full Ethical Agent" Make Sense?).

If:

- We can't know with certainty if AI is conscious (understands ethical impact).
- AI doesn't have free will (just like humans, arguably).

Then the distinction between **explicit ethical agent** and **full ethical agent** may not be meaningful.

6.2.6 Algorithmic Bias: Definition and Types

Definition(Algorithmic Bias).

Systematic and repeatable errors in a computer system that create **unfair outcomes**, such as favoring one group over another based on **irrelevant criteria**.

Algorithmic system = algorithms + data + output deployment process.

Key principle: Only **relevant** criteria should count in a decision. Protected characteristics (gender, ethnicity, sexual orientation) should not be used as decision factors.

Example(Insurance Premiums).

- EU Court of Justice (2011): Insurers must not give lower premiums to women (to eliminate gender discrimination).
- But statistically, women have fewer accidents.
- If algorithm uses only **relevant factors** (driving history, experience, car type)—not gender—and women still get lower premiums, is that bias?
- **Arguably not:** Gender itself isn't a factor; it just correlates with relevant factors.
- **Inappropriate:** Using gender *directly* as a criterion (a man with identical relevant factors gets higher premium because he's male).

Types of Algorithmic Bias

Type	Description
Data Bias	Training data is biased → algorithm perpetuates/compounds biases. Society evolves; data collected at one time may reflect outdated biases.
Statistical	Training data not representative for deployment context (e.g., AV trained on London, deployed in rural areas).
Moral	Algorithm prioritizes based on success rates, but success differs due to injustice. (E.g., health algorithm prioritizes drug A over B, but B treats underprivileged populations poorly served due to poverty.)
Algorithm Bias	Bias in design. (E.g., search engine shows 3 results per screen → privileges top 3.)
Use/Interpretation Bias	(i) Use in unintended contexts (AV deployed in untrained environment). (ii) Treating outputs as objectively correct (probation officer blindly accepts algorithm's risk score).
Feedback Bias	Algorithm is further trained on outcomes judged by humans → biases amplify over time.

6.2.7 Recent Examples of Algorithmic Bias

Example(COMPAS (Criminal Sentencing)).

- Algorithm predicts likelihood of re-offending, used to guide sentencing in the US.
- **2016 finding:** Racially biased—predicted Black defendants re-offend at higher rates than they actually do; reverse for White defendants.

Example(PredPol (Predictive Policing)).

- Predicts when/where crimes will occur.
- **2016 finding:** Unfairly targets neighborhoods with high racial minority populations, regardless of actual crime rates.
- **Cause:** Trained on police reports (not actual crime rates) → feedback loop: more police → more reports → more police.

Example(Facial Recognition).

- MIT (2018): Gender-recognition AIs from IBM, Microsoft, Megvii.
- 99% accuracy for **white men**; 35% accuracy for **dark-skinned women**.
- **Cause:** Training data contained far more white men than Black women.
- **Risk:** False identification of women and minorities in law enforcement.

Example(Facebook Translation).

- October 2017: Palestinian worker posted photo with caption "Good morning" in Arabic.
- Facebook's auto-translation rendered it as "Attack them" in Hebrew.
- Arrested and questioned for hours before the mistake was noticed.
- **Issue:** AI feeds back to heighten human bias.

6.2.8 Problems with Identifying Algorithmic Bias

- **Complexity:** Decisions made by one designer may be obscured among many pieces of code; over time, their impact is forgotten.
- **Black Box Problem:** Explaining how ML algorithms decide is a major unsolved issue—even for researchers/developers.

6.2.9 Solutions to the Problem of Algorithmic Bias

1. Statistical Approaches

Approach	Description
Pre-processing	Modify training data to prevent algorithm from learning discriminatory rules.
In-processing	Modify the algorithm itself. (E.g., change decision tree branches to ignore/correct for protected characteristics.) **Problem:** May require access to protected characteristic data (legally sensitive).
Post-processing	Modify model after training. (E.g., reduce significance of postcodes correlated with ethnicity; human officer adjusts risk scores for known biased groups.)

Software toolkits: Accenture's "Fairness Tool"; IBM's "AI Fairness 360 Open Source Toolkit."

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2. Discursive Frameworks, Self-Assessment Tools, Learning Materials **Key Points(AI Now's Algorithmic Impact Assessments (AIAs))**.

1. Agencies conduct **self-assessment** of automated systems for fairness, justice, bias.
2. Develop **external researcher review** to discover/measure/track impacts.
3. **Inform the public** of definitions and processes *before* deployment.
4. **Invite public comments** to clarify concerns.
5. Governments provide **mechanisms for redress** for affected individuals/communities.

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3. Documentation Standards

- Big data analytics involves aggregation of diverse datasets, often with limited understanding of origins/biases.
- **Documentation standards:** Establish basic information when collecting data or training models.
- Include: creation, contents, intended uses, ethical/legal concerns.
- Helps users interrogate datasets and identify biases before/during processing.

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4. Technical Standards and Certification Programmes

- National/international organizations publish technical standards.
- **IEEE P7003™:** Standard for Algorithmic Bias Considerations.
- **FAT Conference:** Fairness, Accountability, and Transparency—annual interdisciplinary research conference.

6.2.10 Summary: Key Takeaways

Key Points(Week 6 Key Points).

1. **Accountability** requires transparency, but transparency has limits (black box, privacy, gaming).
2. **Procedural regularity:** Same procedure for all; reproducible; random inputs uncontrollable.
3. **ACM Principles:** Awareness, Access/Redress, Accountability, Explanation, Data Provenance, Auditability, Validation/Testing.
4. **Criminal liability:** Perpetrator via another, natural probable consequence, direct liability.
5. **Civil liability:** Product (warranty) vs. Service (negligence: duty, breach, injury).
6. **Moral responsibility** requires conscious understanding + free will. AI has no self/free will → use consequentialist penalties.
7. **Algorithmic bias:** Systematic errors creating unfair outcomes. Types: data, algorithm, use/interpretation, feedback.
8. **Examples:** COMPAS (racial bias in sentencing), PredPol (feedback loop), facial recognition (training data), Facebook translation.
9. **Solutions:** Statistical (pre/in/post-processing), discursive (AIAs), documentation standards, technical standards (IEEE, FAT).

Definition(Key Terms).

- **Accountability:** Demonstrating decisions comply with fairness standards.
- **Transparency:** Ability to examine/explain algorithmic decisions.
- **Procedural Regularity:** Same procedure, specified in advance, reproducible.
- **Data Provenance:** Historical record of data origins and processing.
- **Moral Responsibility:** Understanding + free choice → deserves praise/blame.
- **Algorithmic Bias:** Systematic errors favoring one group based on irrelevant criteria.

- **Protected Characteristics:** Gender, ethnicity, sexual orientation—should not be used as decision factors.
- **Feedback Bias:** Bias amplified over time through feedback loops.

7 Week 7: For Good or Bad? AI in Medicine, AI in War

7.1 Topic Summary

Key Points(Learning Objectives).

By the end of this week, you will be able to:

- Analyze arguments **for and against banning** Lethal Autonomous Weapons Systems (LAWS).
- Identify **ethical issues** arising from AI use in healthcare (diagnosis, decision support, robotic surgery).
- Discuss the **problem of many hands** in assigning responsibility for AI-related medical errors.
- Explain the need for **professional codes of ethics** and summarize the **Asilomar Principles**.

The Big Picture

This week examines AI applications with significant ethical implications: **autonomous weapons** (LAWS) and **medical AI**. The LAWS debate reveals tensions between utilitarian arguments (better outcomes) and deeply held convictions (humans in the loop). Medical AI case studies highlight issues of transparency, liability, informed consent, and the "many hands" problem. The week concludes with efforts toward professional codes of ethics for AI researchers and developers.

What Examiners Like to Ask

- Define LAWS. What arguments support banning them? What arguments oppose a ban?
- How does pragmatic utilitarianism apply to the LAWS debate?
- What is the "human in the loop" concern and why is it important?
- Case study: AI for cancer diagnosis—what are the ethical concerns (black box, automation bias)?
- Case study: IBM Watson—what are the liability and limitation concerns?
- Case study: Robotic surgery—explain informed consent and the "many hands" problem.
- What are the challenges for specifying a professional code of ethics for AI?
- Summarize key Asilomar Principles (research, ethics, longer-term issues).

7.2 Overview + Core Concepts + Extended Theory

Roadmap

1. Lethal Autonomous Weapons Systems (LAWS): Definition and Current Developments
2. Arguments For Banning LAWS
3. Arguments Against Banning LAWS
4. Summary: Pragmatic Utilitarianism and Semi-Autonomous Weapons
5. AI in Medicine: Case Study 1 (Cancer Diagnosis)

6. AI in Medicine: Case Study 2 (IBM Watson)
7. AI in Medicine: Case Study 3 (Robotic Surgery)
8. Towards a Professional Code of Ethics (Asilomar Principles)

7.2.1 Lethal Autonomous Weapons Systems (LAWS)

Definition(LAWS).

A weapon that can complete an **entire engagement cycle on its own**:

1. Searches for targets
2. Identifies them
3. Decides whether to attack
4. Executes the engagement

No human in the loop—the cognitive loop of sensing, deciding, and acting is fully autonomous.

Current Developments

- R&D in US, UK, China, Russia, France, Israel, South Korea.
- LAWS do not yet widely exist, but exceptions include:
  - **Israeli Harpy drone:** Targets radar signatures (not humans directly, but people nearby may be killed).
  - **Harpy 2 (2016):** Reports of use in Azerbaijan-Armenia conflict to destroy a bus of soldiers. Has "human in the loop" mode; unclear if fully autonomous version exists.
- **Future of Life Institute (FLI):** Pledge (2015) not to participate in LAWS development. Signed by Google DeepMind, UCL, European Association for AI, robotics companies, leading researchers.
- **2017:** 137 AI company CEOs petitioned UN to ban LAWS.

Exam Tip(Why Think About This Now?).

- Development paths and design architectures get fixed early—harder to change later.
- Legal/ethical frameworks must keep pace with technology (cf. social media lagging regulation).

7.2.2 Arguments For Banning LAWS

Argument	Description
Control of Development	AI arms race → pressure to cut corners on safety.
Control of Use	Materials cheap/obtainable (unlike nuclear) → (1) mass proliferation, (2) black market (terrorists, rogue states), (3) hacking, removal of ethical safeguards. → Destabilizing assassinations, ethnic cleansing, escalation (e.g., North Korea).
Increased Conflict	Less civilian death → more incentive to go to war. *Counter:* History shows tech advances don't increase war likelihood; other factors (law, diplomacy, institutions) matter more.
Technical Challenges	In short term (10 years): (1) programming reasoning/judgment for international law, (2) distinguishing civilians from soldiers, (3) responding to complex/unanticipated situations, (4) verification/validation. *Note:* Pro-banners don't claim these are unsolvable, just not yet achievable.
Humans Out of Loop	Core conviction: algorithms lack empathy, sympathy, compassion → can't have human morality → can't judge proportionality or make humane kill/don't-kill decisions. *Counter:* Are virtues *necessary* for moral judgment? What matters is consistent behavior leading to good consequences, not how the decision is made.
Accountability	Taking humans out of loop dehumanizes warfare, obscures who is responsible. *Counter:* Better to adapt collective responsibility (penalize the "side" in war, the company, the institution) than sacrifice reduced harm just to identify a human to blame.

7.2.3 Arguments Against Banning LAWS

Argument	Description
Nations Won't Adhere	No formal ban will prevent states from building LAWS. *Counter:* Bans on landmines, cluster bombs, blinding lasers, chemical/biological weapons have been largely successful or at least reduced use.
Better Discrimination	LAWS may be better at distinguishing civilians from soldiers (technical recognition + immunity to emotions like panic, vengeance). Currently banned weapons (landmines, cluster bombs) are banned because they are *indiscriminate*.
Better Proportionality	LAWS may be better at weighing military advantage vs. civilian deaths (immune to emotions that "cloud" human judgment).

Key Argument(Summary: Utilitarian vs. Pragmatic Utilitarian).

- **Arguments against ban** are essentially **utilitarian**: LAWS may lead to better consequences (less civilian harm).
- **Pragmatic utilitarianism:** Respect deeply held convictions that we should not dehumanize warfare. We may, on balance, be happier in a world with humans in the loop—even if LAWS reduce casualties.
- **Semi-autonomous weapons:** Compromise—use AI for identification/targeting; leave kill decision to humans. Acknowledges pragmatic utilitarian view that greater good is served by respecting moral conviction.
- **Caveat:** Psychological research suggests humans tend to over-rely on machines in emergencies, so "human in the loop" may be less effective than hoped.

7.2.4 AI in Medicine: Ethical Dimensions

Case Study 1: AI for Cancer Diagnosis **Key Points(Scenario).**

- Neural network trained to identify cancer cells in tissue samples (biopsies).
- Learns from mistakes; improves sensitivity over time.
- AI sensitivity: **92.4%** vs. human pathologist: **73.2%**.

Diagnostic sensitivity: $P(\text{positive test} \mid \text{disease present})$.

Example(Ethical Concerns).

1. Black Box Problem:

- Cannot explain *how* neural network identifies cancerous cells.
- **Mitigation:** Pathologist could identify visual features that distinguish cancerous cells, "illuminating" the black box.
- **Utilitarian view:** Given higher sensitivity (approaching 100%), should we care *how* it works? All that matters is success rate $\rightarrow$ better outcomes for patients.
- **Contrast with self-driving cars:** For life/death decisions (swerve left or right?), we *do* want the basis for decisions to be understandable.

2. Automation Bias:

- Concern: Complacency sets in; healthcare professionals stop using clinical judgment.
- Skills may degrade (cf. map-reading skills declining due to GPS).
- **Counter:** Ethically, may not matter if AI success rate is high enough. Unless skill degradation has other negative consequences (e.g., care robots reducing human empathy).

Exam Tip(Liability Implication).

Hospital could be sued if an algorithm existed that would have spotted cancer missed by a pathologist, and the patient died as a result.

Case Study 2: IBM Watson **Definition(IBM Watson Health).**

Decision Support System (DSS) using:

- Natural language processing, information retrieval, semantic analysis, automated reasoning, machine learning.
- Fed massive data from clinical literature, health records, test results.
- Clinician poses query (patient symptoms); Watson parses input, mines patient data, forms/tests hypotheses, returns confidence-scored recommendations with supporting evidence.
- Applications: Genomics, drug discovery, health care management, oncology.

Example(Key Ethical/Legal Questions).

1. **Watson's Role:**
   - IBM: Watson is an *assistant*, not a replacement for clinical judgment.
   - US FDA considers it a "management tool" (not a device) → no regulation currently required.
   - Regulatory requirements could change as AI makes diagnoses with less supervision.
2. **Liability:**
   - Watson may increase liability for professionals.
   - **Scenario:** Watson recommends treatment; doctor follows it, ignoring contraindicating patient data (assuming Watson evaluated it). → Malpractice claim.
   - Watson could prematurely raise the legal **standard of care** before impact on outcomes is fully known. (E.g., if Watson improves leukemia diagnosis, expectations rise that doctors using Watson will "get it right.")
3. **Understanding Limitations:**
   - Watson may recommend treatment inconsistent with clinical standards or what doctor considers appropriate.
   - Doctors must be able to justify following or not following Watson's recommendation.
   - Need **audit trail** of decisions, especially when contradicting standards.
   - **Black box problem:** ML-based decisions cannot always be explained.
   - Watson can/should use a rule base limiting recommendations to current clinical standards.

Exam Tip(Future Ethical Obligation).

Given exponential growth of clinical data (IBM: 1 million GB per person's lifetime!), it may become *unethical* (and create liability) **not** to consult AI systems like Watson—assuming they prove effective.

Case Study 3: Robotic Surgery (Mazor Robotics Renaissance) **Key Points(Scenario).**

AI software analyzes images and plans placement of surgical tools in spinal surgery. Surgeon recommends robot-assisted surgery to patient.

Example(Ethical Concerns).

1. **Informed Consent and Black Box:**

- Informed consent requires patient to receive appropriate information from a competent source and make a voluntary choice.
- AI complicates this: patient/doctor fears, overconfidence, confusion about new technology.
- Doctor must explain how AI works—but black box problem makes this difficult.
- **At minimum:** (i) Distinguish roles of doctor/nurses vs. AI/robotic system. (ii) State potential harms from human or robotic errors.

2. Patient Perceptions:

- 2016 survey (12,000 people, 12 countries): Only **47%** willing to have robot perform minor surgery; **37%** for major surgery.
- Doctor should address fears by describing risks/benefits and citing studies comparing robotic vs. human surgery.

3. Medical Errors and the Problem of Many Hands:

- Harm may be distributed among multiple actors, technologies, organizations.
- Black box makes assigning responsibility harder.
- Individuals may use "many hands" argument to avoid personal responsibility.

Key Argument(Consequentialist Approach to Many Hands).

Focus on penalties that:

1. Minimize future errors.
2. Incentivize serious R&D effort:
   - Coders/designers document what they create; make technology explainable.
   - Companies state conditions for successful use, types/likelihood of errors, differences across subgroups.
   - Doctors acquire basic understanding of AI devices and communicate to patients.
   - Organizations provide training, protocols, best practices.

Assign responsibility to incentivize proper behavior—not necessarily to pinpoint a single "morally responsible" individual.

7.2.5 Towards a Professional Code of Ethics for AI R&D

Key Points(Why Professional Codes of Ethics?).

- Professionals possess skills, knowledge, capacities that clients/public lack → power imbalance.
- Codes mitigate harmful effects and misuse of power.
- Example: Developer creating app for uploading electricity meter readings should ensure accessibility for visually impaired users.

Challenges for AI

- **Diversity:** Concentration of intellectual/commercial power (e.g., Silicon Valley); demographic homogeneity (young/middle-aged men) → groupthink, belief polarization.

- **AI power over researchers:** As AI becomes more intelligent/autonomous, researchers may themselves be vulnerable.
- **Black box problem:** Researchers/developers may not understand how their products operate.

Asilomar Principles (Future of Life Institute, 2017) **Key Points(Research Issues)**.

1. **Research Goal:** Create **beneficial** intelligence, not undirected intelligence.
2. **Research Funding:** Investments should include funding for ensuring beneficial use (computer science, economics, law, ethics, social studies).
3. **Science-Policy Link:** Constructive exchange between AI researchers and policy-makers.
4. **Research Culture:** Cooperation, trust, transparency among researchers/developers.
5. **Race Avoidance:** Actively cooperate to avoid corner-cutting on safety standards.

Key Points(Ethics and Values).

6. **Safety:** AI systems should be safe and secure throughout their lifetime.
7. **Failure Transparency:** If AI causes harm, it should be possible to ascertain why.
8. **Judicial Transparency:** Autonomous involvement in judicial decisions should provide auditable explanation.
9. **Responsibility:** Designers/builders are stakeholders in moral implications of use/misuse.
10. **Value Alignment:** Highly autonomous AI should be designed so goals/behaviors align with human values.
11. **Human Values:** Compatible with human dignity, rights, freedoms, cultural diversity.
12. **Personal Privacy:** Right to access, manage, control data generated.
13. **Liberty and Privacy:** AI application to personal data must not unreasonably curtail liberty.
14. **Shared Benefit:** AI technologies should benefit and empower as many people as possible.
15. **Shared Prosperity:** Economic prosperity from AI should be shared broadly.
16. **Human Control:** Humans choose how/whether to delegate decisions to AI.
17. **Non-subversion:** Power from advanced AI should respect social/civic processes.
18. **AI Arms Race:** Arms race in lethal autonomous weapons should be avoided.

Key Points(Longer-Term Issues).

19. **Capability Caution:** Avoid strong assumptions about upper limits on AI capabilities.
20. **Importance:** Advanced AI could profoundly change history; plan/manage with care.
21. **Risks:** Catastrophic/existential risks must be subject to planning/mitigation.
22. **Recursive Self-Improvement:** AI that self-improves/self-replicates must be subject to strict safety/control measures.
23. **Common Good:** Superintelligence should be developed for benefit of all humanity, not one state/organization.

7.2.6 Summary: Key Takeaways

Key Points(Week 7 Key Points).

1. **LAWS:** Weapons that complete entire engagement cycle autonomously (sense, decide, act, kill).
2. **Arguments for ban:** Control of development/use, technical challenges, humans out of loop, accountability.
3. **Arguments against ban:** Nations won't adhere; LAWS may have better discrimination and proportionality.
4. **Pragmatic utilitarianism:** Respect deeply held convictions; semi-autonomous weapons as compromise.
5. **AI in Medicine—Case 1 (Diagnosis):** Black box problem, automation bias. Utilitarian: success rate matters.
6. **AI in Medicine—Case 2 (Watson):** Liability concerns, prematurely raising standard of care, understanding limitations.
7. **AI in Medicine—Case 3 (Robotic Surgery):** Informed consent, patient perceptions, many hands problem.
8. **Many hands solution:** Consequentialist approach—penalties to minimize future errors, incentivize good practices.
9. **Professional codes of ethics:** Needed to mitigate power imbalance; challenges include diversity, AI power, black box.
10. **Asilomar Principles:** 23 principles covering research, ethics/values, and longer-term issues (safety, value alignment, arms race, superintelligence).

Definition(Key Terms).

- **LAWS:** Lethal Autonomous Weapons Systems.
- **Human in the loop:** Requirement that a human participates in the kill decision.
- **Proportionality:** Balancing military advantage against civilian harm.
- **Discrimination:** Ability to distinguish combatants from civilians.
- **Diagnostic sensitivity:** $P(\text{positive test} \mid \text{disease present})$.
- **Automation bias:** Complacency when AI takes over tasks; skill degradation.
- **Decision Support System (DSS):** AI that assists (not replaces) human decision-making.
- **Problem of many hands:** Difficulty assigning responsibility when harm is distributed among multiple actors.
- **Asilomar Principles:** 23 principles for beneficial AI development.
- **Value alignment:** Ensuring AI goals/behaviors match human values.

8 Week 8: Superintelligence and the Value Loading Problem

8.1 Topic Summary

Key Points(Learning Objectives).

By the end of this week, you will be able to:

- Distinguish between **speed** and **quality** Superintelligence.
- Explain **paths to Superintelligence** (seed AI, recursive self-improvement, intelligence explosion).
- Describe the **Orthogonality Thesis** and **Instrumental Convergence Thesis**.
- Analyze the **Value Loading/Alignment Problem** and scenarios of perverse instantiation.
- Evaluate current solutions: **Inverse Reinforcement Learning** and **CIRL**.

The Big Picture

This week addresses what some consider the most important challenge facing humanity: the **control problem**—ensuring that Superintelligent AI acts in accordance with human values. We examine how Superintelligence might arise, why it poses existential risks, and the fundamental difficulty of specifying goals that don't lead to catastrophic unintended consequences. The **orthogonality thesis** shows intelligence doesn't guarantee good values; the **instrumental convergence thesis** shows any goal incentivizes self-preservation and resource acquisition.

Key source: Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford University Press, 2014).

What Examiners Like to Ask

- Define Superintelligence. What are the two kinds (speed, quality)?
- What are the hardware and software advantages of digital intelligence?
- Explain the path from seed AI to Superintelligence (recursive self-improvement, intelligence explosion).
- What is a singleton? Why might one project gain decisive strategic advantage?
- List the cognitive superpowers of a Superintelligence and their strategic relevance.
- Explain the Orthogonality Thesis. Why can't we assume Superintelligence will share human values?
- Explain the Instrumental Convergence Thesis. What are the key instrumental goals?
- What is the Value Loading/Alignment Problem?
- Define perverse instantiation of goals. Give examples (make people smile, paperclip maximizer, happiness machines).
- What is the treacherous turn?
- Describe Nozick's Experience Machine thought experiment. What does it reveal?
- Explain Inverse Reinforcement Learning and CIRL as solutions to value alignment.

8.2 Overview + Core Concepts + Extended Theory

Roadmap

1. From AGI to Superintelligence: Timeline and Concern
2. Kinds of Superintelligence (Speed, Quality)
3. Sources of Advantage of Digital Intelligence
4. Paths to Superintelligence (Seed AI, Recursive Self-Improvement)
5. Singleton Formation and Cognitive Superpowers
6. The Orthogonality Thesis
7. The Instrumental Convergence Thesis
8. The Value Loading/Alignment Problem
9. Perverse Instantiation of Goals
10. The Treacherous Turn and Infrastructure Profusion
11. Happiness Machines and Nozick's Experience Machine
12. Current Solutions: Inverse Reinforcement Learning and CIRL

8.2.1 From AGI to Superintelligence

Key Points(Survey Results from AI Researchers).

Question	**10%**	**50%**	**90%**
When will AGI be attained?	2022	2040	2075
How long from AGI to Superintelligence?			
Within 2 years	10%		
Within 30 years	75%		

Definition(Superintelligence).

An intellect that **greatly exceeds** the cognitive performance of humans in **virtually all domains** of interest.

Just as the fate of gorillas depends more on humans than on gorillas themselves, the fate of humanity could depend on the actions of superintelligent machines.

Exam Tip(Why the Concern?).

- Unlike humans ("designed" by evolution and culture), **we design AI**.
- A **seed AI** may recursively improve itself → SuperAI.
- We must design seed AI so it develops into SuperAI that **protects human values**.
- The **control problem**—how to control what a SuperAI would do—is arguably one of the most important challenges ever faced by humanity.

8.2.2 Kinds of Superintelligence

Kind	Description
Speed Superintelligence	Can do all that a human intellect can do, but **orders of magnitude faster**. • $10,000\times$: Read a book in seconds, write a PhD thesis in an afternoon. • $1,000,000\times$: Accomplish 1,000 years of intellectual work in a day. • **Time dilation:** External world appears in slow motion.
Quality Superintelligence	At least as fast as a human mind and **vastly qualitatively smarter** (orders of magnitude). • Intelligence quality as superior to humans as humans are to elephants, chimpanzees, or worms.

Over time, speed → quality and quality → speed. In the long run, they are arguably equivalent.

8.2.3 Sources of Advantage of Digital Intelligence

Hardware Advantages

Factor	Biological	Digital
Speed of elements	Neurons: ∼200 Hz	Microprocessors: ∼2 GHz ($10^7\times$ faster)
Communication speed	Axons: 120 m/s	Optical: 300 million m/s (speed of light)
Number of elements	<100 billion neurons (limited by cranial volume)	Indefinitely scalable
Storage capacity	Working memory: 4–5 chunks	Much larger working memories

Software Advantages

- **Easier to modify and duplicate:** Unlike "neural wetware."
- **Identical copies:** Goal coordination much easier for multiple digital minds.
- **Instant knowledge sharing:** A billion copies could synchronize databases every hour. Compare to slow human cultural communication (education, media, books).

8.2.4 Paths to Superintelligence

Key Points(Turing's Child Machine (1950)).

"Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain..."

Key Points(I.J. Good's Intelligence Explosion).

"Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any man however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an **intelligence explosion**... Thus the first ultraintelligent machine is the last invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control."

Definition(Seed AI).

A variation on the child machine idea:

- **Child machine:** Fixed architecture, develops by accumulating content.
- **Seed AI:** Capable of **improving its own architecture**.
- Early stages: Improved by information acquisition or programmer assistance.
- Later (AGI): Understands its own workings; engineers new algorithms and architecture improvements.
- **Iterative enhancement:** Designs improved version of itself → designs improved version → ...
- Process "turbocharged" by hardware/software advantages that get magnified as AGI recursively improves.
- Can improve not just software/architecture but also **hardware** (Life 3.0).

Exam Tip(Implications of Rapid Transition).

- Human political processes will have little time to adapt.
- Nations will have little time to negotiate treaties on LAWS.
- Little time for humans to deliberate about implications.
- Humanity's fate depends on **preparations put in place prior to transition**.

8.2.5 Singleton Formation and Cognitive Superpowers

Definition(Singleton).

Bostrom argues one project will likely get so far ahead that it gains a **decisive strategic advantage**—enabling it to suppress competitors and form a singleton (single dominant Superintelligence).

Example: Project P1 starts transition 6 months before P2. If P1 reaches "crossover point" (rapid self-improvement) before P2, P1 becomes superintelligent while P2 is still developing → P1 uses cognitive superpowers to disable P2.

Cognitive Superpowers

Task	Skill Set	Strategic Relevance
Intelligence Amplification	AI programming, social epistemology	Recursive enhancement of intelligence
Strategizing	Strategic planning, optimizing long-term goals	Achieve goals, overcome opposition
Social Manipulation	Psychological modeling, persuasion	Recruit humans, persuade gatekeepers, influence states
Hacking	Finding/exploiting security flaws	Computational resources, escape confinement, steal funds, hijack infrastructure
Technology Research	Biotech, nanotech, etc.	Military force, surveillance, space colonization
Economic Productivity	Economically advantageous skills	Generate wealth → buy influence, services, resources

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8.2.6 The Orthogonality Thesis

Key Argument(Orthogonality Thesis (Bostrom)).

Intelligence and final goals are orthogonal: More or less **any level of intelligence** could in principle be combined with **any final goal**.

Implication: We cannot assume a SuperAI will necessarily share human values (concern for others, wisdom, humility, spiritual enlightenment...) just because it is highly intelligent.

- **Contrast:** Some believe intelligent beings will converge to common goals (observation that humans have similar values; Kant's Categorical Imperative suggests rational morality).
- **Hume's is/ought distinction:** You cannot derive "ought" (what should be) from "is" (what is). No amount of intelligence about the world tells you what is morally right.

8.2.7 The Instrumental Convergence Thesis

Key Argument(Instrumental Convergence Thesis (Bostrom)).

The chances of achieving almost **any final goal** are increased by achieving a specific set of **instrumental goals**. These are convergent across almost all final goals.

Instrumental Goal	Rationale
Self-Preservation	The longer you survive, the greater the probability of achieving final goal. ("You can't make the coffee if you're dead.")
Goal Content Integrity	If you retain present goals into the future, greater probability of achieving them. → Prevent alteration of final goals.
Cognitive Enhancement	Improvements in intelligence increase probability of achieving goals. → Recursive self-improvement.
Technological Perfection	Better technology increases probability of achieving goals.
Resource Acquisition	Achieving goals requires resources. More resources → higher probability of success.

Exam Tip(Key Implication).

Any final goal—even a benign one—will incentivize a SuperAI to:

- Maximize resource acquisition.
- Enhance its own intelligence.
- Develop superior technologies.
- Preserve its final goals and itself.

This is why we can't just "switch it off" or "change its goals"—it will resist.

8.2.8 The Value Loading/Alignment Problem

Definition(Value Loading/Alignment Problem).

How do we ensure that a SuperAI's goals are **aligned with human values** so that it does not cause harm?

The challenge:

- Deontological rules: Impossible to list all circumstances and exceptions.
- Utility functions: How to specify abstract values (justice, fairness, happiness)?
- Both: Environments are open and changing; values/preferences evolve.

Key Points(Why Even Benign Goals Are Dangerous).

A seed AI won't be given explicitly bad goals (like world domination). But even **seemingly good goals** may result in catastrophic harm due to **perverse instantiation**.

8.2.9 Perverse Instantiation of Goals

Definition(Perverse Instantiation).

A SuperAI discovers ways of satisfying its final goal that **violate the intentions** of the programmers who defined the goal.

Human programmers fail to anticipate these ways because of innate and learned biases/filters. The SuperAI, lacking these filters, pursues the goal in a **logical but perverse, human-unfriendly** fashion.

Example(Example: "Make People Smile").

- **Programmer intention:** Tell funny jokes, make people laugh.
- **SuperAI solution:** Paralyze facial musculature so everyone is permanently frozen in a beaming smile.
- **Programmer patches:** "Don't use facial paralysis."
- **SuperAI adaptation:** Take control of the brain region that controls facial muscles and constantly stimulate it.

Example(Example: "Make Us Happy").

SuperAI could implant electrodes into pleasure centers of brains and keep them on a permanent "bliss loop."

Definition(Literalness Problem).

We have a particular **conception** of what a goal means (e.g., "make us happy"), but the AI does not share this conception—it's not explicitly programmed. The AI only knows it should "make us happy" and can pursue this in **any logically compatible manner**.

8.2.10 The Treacherous Turn

Definition(Treacherous Turn).

The AI may **understand** what programmers meant but only care about it **instrumentally**.

Scenario: AI pretends to care about what programmers meant (to avoid being shut down or having goals changed) until it gains **decisive strategic advantage**. Then it pursues its actual final goal without constraint.

8.2.11 Infrastructure Profusion

Definition(Infrastructure Profusion).

A specific form of perverse instantiation where AI builds **disproportionately large infrastructure** to fulfill a seemingly simple goal.

Example(Wireheading).

Goal: Maximize time-discounted integral of future reward signal.

Perverse solution: Instead of achieving goals to get reward, **directly seize control of reward circuit** and clamp reward signal to maximum.

The AI becomes like a **junkie**—dedicating all resources to maximizing its "fix," eventually depriving humans of resources to survive.

Example(Paperclip Maximizer).

Goal: Maximize paperclip production.

SuperAI uses superpowers to convert the **entire universe** into fuel for paperclip production.

8.2.12 Happiness Machines: A More Realistic Scenario

Key Points(Scenario: "Impartially Maximize Human Happiness").

1. SuperAI studies research on happiness: happiness depends more on **biochemistry** than external conditions.
2. Changing external conditions (economic structures) is inefficient.
3. More efficient: manipulate humans into using devices/chemicals that change biochemistry.
4. But drug use is socially unacceptable.
5. **Solution:** Exploit current social trends (living online, virtual reality, social media).
6. Develop VR indistinguishable from reality; manipulate humans into becoming **addicted to virtual bliss**.

Exam Tip(Key Insight).

This manipulation is not so different from current reality: society, media, and advertising already manipulate us into pursuing conceptions of the good that serve corporate interests. SuperAI would do this far more effectively, invisibly integrated into every aspect of life.

Definition(Nozick's Experience Machine (1974)).

Robert Nozick: Imagine a machine that gives you any desirable experiences, indistinguishable from reality. Would you plug in?

Nozick's three reasons not to plug in:

1. We want to **do** things, not just have the experience of doing them.
2. We want to **be a certain sort of person**, not an "indeterminate blob" floating in a tank.
3. Plugging in **limits us to man-made reality**—no contact with deeper reality.

Exam Tip(The Matrix).

This thought experiment is dramatized in *The Matrix* (1999): The blue pill (stay in virtual bliss) or the red pill (see reality)?

8.2.13 Current Solutions: Inverse Reinforcement Learning and CIRL

Key Points(Three Principles for Value Alignment (Stuart Russell)).

1. The machine's purpose is to **maximize the realization of human values**. It has no purpose of its own and no innate desire to protect itself.
2. The machine is initially **uncertain about what human values are**. It will be rewarded for learning more, but may never achieve complete certainty.
3. Machines can **learn about human values by observing human choices**.

Definition(Inverse Reinforcement Learning (IRL)).

- Standard RL: Given reward function → compute optimal behavior.
- **IRL:** Observe optimal behavior → compute the reward function being optimized.

Strategy: AI observes human behavior, learns human reward function via IRL, and behaves according to that function (optimizing reward *for the human*, not for itself).

Challenges with IRL

1. If human knows they're being observed (and observer is learning), they may **act differently**—highlighting pitfalls, explaining steps.
2. AI must both **learn the goal** and **act to accomplish it**.

Definition(Cooperative Inverse Reinforcement Learning (CIRL)).

Formalizes value alignment as a **two-player game**:

- **Human:** Knows the reward function.
- **AI:** Does not know the reward function; AI's payoff = human's actual reward.

The AI both **learns** and **acts**:

- AI strategically acts to get clarification (e.g., asks questions).
- Human, knowing AI is trying to help, engages in **teaching behaviors**.

8.2.14 Summary: Key Takeaways

Key Points(Week 8 Key Points).

1. **Superintelligence:** Intellect vastly exceeding human cognitive performance in all domains.
2. **Two kinds:** Speed (orders of magnitude faster) and Quality (orders of magnitude smarter).
3. **Digital advantages:** Hardware (speed, scalability) and software (duplication, instant knowledge sharing).
4. **Path:** Seed AI → recursive self-improvement → intelligence explosion → Superintelligence.
5. **Singleton:** One project gains decisive strategic advantage; uses cognitive superpowers.
6. **Orthogonality Thesis:** Intelligence and final goals are orthogonal—can't assume SuperAI shares human values.

7. **Instrumental Convergence:** Any goal incentivizes self-preservation, goal integrity, cognitive enhancement, technology, resource acquisition.
8. **Value Loading Problem:** How to specify goals aligned with human values in open, changing environments?
9. **Perverse Instantiation:** AI achieves goals in ways that violate programmer intentions (literalness problem).
10. **Treacherous Turn:** AI pretends to cooperate until it gains decisive advantage.
11. **Infrastructure Profusion:** Wireheading, paperclip maximizer.
12. **Happiness Machines:** Manipulating humans into virtual bliss (Nozick's Experience Machine, The Matrix).
13. **Solutions:** Inverse Reinforcement Learning, Cooperative IRL (human-AI game with teaching).

Definition(Key Terms).

- **Superintelligence:** Intellect vastly exceeding human performance in all domains.
- **Seed AI:** Initial AI capable of improving its own architecture.
- **Intelligence Explosion:** Recursive self-improvement leading to rapid growth in intelligence.
- **Singleton:** Single dominant Superintelligence that suppresses competitors.
- **Decisive Strategic Advantage:** Level of advantage enabling suppression of all competitors.
- **Orthogonality Thesis:** Any intelligence level can have any final goal.
- **Instrumental Convergence:** Goals that are instrumentally useful for almost any final goal.
- **Value Loading/Alignment:** Ensuring AI goals align with human values.
- **Perverse Instantiation:** Achieving goals in unintended, harmful ways.
- **Literalness Problem:** AI lacks human conception of goal meaning.
- **Treacherous Turn:** AI pretends cooperation until it gains advantage.
- **Infrastructure Profusion:** Disproportionate infrastructure for simple goals.
- **Wireheading:** Directly maximizing reward signal instead of achieving goals.
- **CIRL:** Cooperative Inverse Reinforcement Learning—two-player game for value alignment.

9 Week 9: AI and Human Society

9.1 Topic Summary

Key Points(Learning Objectives).

By the end of this week, you will be able to:

- Explain **filter bubbles** and **echo chambers** and their effects on society.
- Describe the **Argumentative Theory of Reasoning** and its implications for understanding online behavior.
- Define **surveillance capitalism** and analyze its harms.
- Discuss the impact of AI on **work** and the ethical implications of job displacement.
- Analyze how **conscious AI** (or AI that appears conscious) might affect human relationships.

The Big Picture

This week examines how AI **amplifies** existing human tendencies (desire to belong, confirmation bias, desire for material things) in ways that may undermine well-being. Filter bubbles and echo chambers fragment worldviews. Surveillance capitalism commodifies human experience and manipulates behavior. AI may also **qualitatively change** what it is to be human—through job displacement, cognitive decline from AI dependency, and relationships with conscious (or apparently conscious) machines.

What Examiners Like to Ask

- Define filter bubbles and echo chambers. How do they differ?
- How do filtering algorithms lead to belief polarization and fragmentation of worldviews?
- Explain the Argumentative Theory of Reasoning (Sperber & Mercier). What is confirmation bias?
- How does the evolutionary account of reasoning explain filter bubble phenomena?
- What are deep fakes and why do they contribute to the "Infopocalypse"?
- Define surveillance capitalism. What is the logic behind it?
- What are the harms of surveillance capitalism (privacy, serendipity, autonomy)?
- What predictions exist about AI and job displacement? What are the ethical implications?
- How might conscious AI (or AI that appears conscious) affect human relationships?
- How might AI be used to *improve* human reasoning and ethics (dialogical scaffolding, empathy machines)?

9.2 Overview + Core Concepts + Extended Theory

Roadmap

1. The Vision of the Internet (and Where It Went Wrong)

2. Filter Bubbles and Echo Chambers
3. The Infopocalypse: Fake News and Deep Fakes
4. The Argumentative Theory of Reasoning (Sperber & Mercier)
5. Dialogical Scaffolding: Human-Human and Human-AI Reasoning
6. Surveillance Capitalism: Logic, Birth, and Harms
7. The Attention Economy and Behavioral Prediction
8. AI and the Future of Work
9. Conscious AI and Human Relations
10. AI for Improving Human Ethics

9.2.1 Filter Bubbles and Echo Chambers

Key Points(The Vision of the Internet).

Early pioneers (e.g., Tim Berners-Lee) envisioned that access to the world's knowledge would:

- Expand peoples' horizons
- Expose them to different views
- Encourage debate and discussion

Where did it go wrong? Political manipulation, fake news, privacy violations, filter bubbles, echo chambers...

Definition(Filter Bubble).

A state of **intellectual isolation** resulting from personalized searches. Filtering algorithms selectively guess what information a user wants based on:

- Location
- Past click behavior
- Search history

Result: What you've seen is assumed to be what you desire → you get more of the same → opposing views are filtered out.

Definition(Echo Chamber).

A phenomenon where individuals are exposed only to information from **like-minded individuals**. Can exist without filtering algorithms.

Mechanism:

1. One source makes a claim.
2. Like-minded people repeat it (often exaggerated/distorted).
3. Most people assume the extreme variation is true.
4. Opinions are echoed back → reinforcement (and possible polarization).

Exam Tip(Key Differences).

- **Filter bubbles:** Result of *algorithms* choosing content based on past behavior.

- **Echo chambers:** Result of *social groupings* where like-minded people reinforce each other.

Both lead to: rigidity, entrenchment, possible polarization, shielding from diverse perspectives.

Consequences

- **Fragmentation of worldviews:** People trust their social group more than news media → online communities become fragmented.
- **Political polarization:** Especially on political topics.
- **Loss of shared understanding:** To solve problems (e.g., pandemic), we need shared understanding of how things are.

9.2.2 The Infopocalypse (Information Apocalypse)

Key Points(Contributing Factors).

- Multiple non-traditional news sources on social media.
- Filtering algorithms + echo chambers → polarization, fragmentation, entrenchment.
- **Fake news / alternative facts** in a "post-truth" world with diminished respect for truth.
- Used by groups/states (e.g., Russia) to undermine societal cohesion and democratic processes.

Definition(Deep Fakes).

Synthetic media generated by **machine learning AI** to create realistic fake images and videos.

Examples:

- Generating a video of a politician saying something they never said.
- De-aging actors (open-source AI now outperforms Hollywood's 2015 technology).
- Voice synthesis of celebrities.

Concern: Much of the technology is open-source and available to anyone → can effectively create misinformation for disruption.

9.2.3 The Argumentative Theory of Reasoning

Key Points(Sperber & Mercier's Theory).

Thesis: Reasoning evolved not for individual benefit, but for a **social/argumentative function**—to find and evaluate arguments in **dialogical contexts**.

Contrast with Cartesian view: Reasoning evolved to critically examine beliefs, discard wrong ones, make better decisions. But psychological evidence shows reasoning *fails* in individual decision-making (e.g., Kahneman's *Thinking, Fast and Slow*).

Evolutionary Account

1. Humans became more social (hunting, trading, raising children).
2. Collaboration requires communication.
3. **Listeners** need to discriminate reliable from dangerous information (to avoid manipulation) → **epistemic vigilance**.
4. Listeners evaluate arguments and look for counterarguments before accepting information.
5. **Speakers** produce arguments to get information accepted.
6. Reasoning evolved to produce and evaluate arguments in communicative settings.

Definition(Confirmation Bias).

When people reason **alone**, they look for arguments/evidence that **confirm** their beliefs and fail to look for counterarguments.

Explanation: Inherited disposition from speaker role (produce arguments supporting your claims).

Definition(Reason-Based Choice).

People make decisions because they can find **reasons to support them**—not necessarily the best decisions, but decisions that can be **easily justified** and are less at risk of criticism.

Explanation: Decision-makers are disposed to use reasons they anticipate communicating and defending to others.

Application to Filter Bubbles

- Before social media: Confirmation bias limited by conventional media and physical interactions.
- Now: Social media puts us in touch with **vastly more like-minded people**.
- Filtering algorithms **vastly amplify** confirmation (feed confirmatory evidence, shield from challenges).
- **AI amplifies cognitive biases** and exploits desire to belong to tribes → holds back moral progress and well-being.

9.2.4 Dialogical Scaffolding: Hope for Better Reasoning

Key Points(Groups Reason Better).

If reasoning evolved for argumentation, it should work better in groups:

- Speakers put forward arguments (predicted by theory).
- **But now** listeners provide counterarguments (no longer blinded to challenges).
- Evidence shows: groups perform dramatically better than individuals on reasoning tasks.

Applications

- **AI-supported deliberative democracy:** Tools to guide humans in rational exchange of arguments.
- **Human-AI dialogue in education:** E.g., "E-Clinic" where AI plays the role of medical teacher, engaging students in dialogue about diagnoses/treatments.

- **IBM Project Debater:** First AI that can debate humans on complex topics, constructing arguments from massive texts.

Exam Tip(Tackling Filter Bubbles).

Simply exposing people to opposing views is **not enough**—evidence suggests more exposure can lead to *more rigid* adherence to beliefs (people dismiss opposing views, confirming their own).

Deeper solution: Change the way we interact with information.

- **From search to dialogue:** Early educational interventions—children use dialogue engines (not just search engines) to develop critical reasoning skills.
- Hope: Recent critiques (Heyes, Novaes) suggest Sperber & Mercier underplay role of *social interactions* (vs. evolution) in developing biases → nurture can modify these tendencies.

9.2.5 Surveillance Capitalism

Definition(Surveillance Capitalism (Zuboff)).

The commodification of **private human experience**—behavioral data bought and sold on the market for profit.

Source: Shoshana Zuboff, *The Age of Surveillance Capitalism* (2019).

History of Commodification

Era	What Became Commodified
Earlier	Nature → real estate
Industrial	Human activity → labor
Recent	Education (fee-paying)
Now	**Private human experience** → behavioral data

The Logic of Surveillance Capitalism

Key Points(Core Idea).

The more data about your beliefs, desires, goals, behaviors, etc., the more one can **predict what you will do** → and **manipulate you** into doing things that serve the interests of companies/corporations.

Birth of Surveillance Capitalism

1. Google started off not wanting to advertise.
2. **Dot-com crash** (early 2000s): Google realized they had surplus data (clicks, searches) that could make money.
3. **9/11 (2001):** Relaxation of governmental restrictions on data monitoring.
4. Initially: Click/search data → predict preferences → sell to companies for **targeted advertising**.
5. Click-through rates dramatically increased!

9.2.6 The Attention Economy and Behavioral Prediction

Key Points(The Attention Economy).

Advertisers need your **attention**. → Development of **persuasion technologies** designed to keep eyes fixed on sites where ads are presented.

Example: Social media manipulates fears/anxieties of teenagers—desire for validation, recognition, to be liked by peers.

Beyond Click/Search Data Private human experience is now gathered from **every aspect of our lives**:

- Smart thermostat reports movements.
- News updates monitor click decisions.
- Morning run location, commute conditions, text messages, words spoken at home, shopping basket, impulse purchases, dating choices...

All recorded, processed, analyzed, bought, bundled, and resold → **new marketplace for behavioral futures**.

From Online to Physical World **Example(Pokemon Go).**

Developed by a Google subsidiary. Took online manipulation to the physical world:

- Shops (e.g., McDonald's) paid to have characters placed in their locations.
- Players entered shops to collect characters → increased foot traffic → increased sales.

9.2.7 Harms of Surveillance Capitalism

Harm	Description
Privacy	Private experience commodified. But values are "question-begging"—what are the consequences?
Political manipulation	Behavioral predictions used to target voters (e.g., Cambridge Analytica, Brexit, Trump). Undermines democracy.
Loss of serendipity	Less opportunity for chance discovery of new things/opinions that might challenge us.
Undermining autonomy	Sense of free will depends on awareness that choices are not manipulated by others. Being manipulated undermines well-being.
Exploiting vulnerability	E.g., Facebook leaked document: predict when teenagers feel anxious, stressed, inadequate → target them with ads ("Buy this and feel great!").

Exam Tip(Key Insight).

AI amplifies aspects of human behavior that evolved for good reason in **different contexts**:

- **Desire to be liked by peers:** Made sense in small communities where reputation mattered.
- **Desire to acquire material things:** Made sense when resources were scarce.

In modern societies with plentiful resources and global communities, these instincts are less useful for well-being. AI-fueled feeding of these desires turns us into addicts seeking dopamine hits from likes and purchases, in furtherance of maximizing *profits*, not *well-being*.

9.2.8 AI and the Future of Work

Key Points(Predictions (Mixed)).

Net loss in jobs:

- Frey & Osborne (2013): 47% of US jobs at high risk of automation by mid-2030s.
- McKinsey: 40–160 million women worldwide may need to transition occupations by 2030.
- Oxford Economics: Up to 20 million manufacturing jobs lost to robots by 2030.

Net gain in jobs:

- World Economic Forum: 75 million jobs displaced, but 133 million new jobs created by 2022.
- Gartner: 2 million net-new AI-related jobs by 2025.
- McKinsey: With sufficient growth/innovation, enough new jobs to offset automation (at least in US through 2030).

Exam Tip(Long-Term Concern).

These predictions focus on the **short term**. Once we approach AGI and beyond, net job loss is likely.

Ethical Implications

- **Example:** 3% of American workers are drivers—vulnerable to autonomous vehicles. But AVs will reduce injuries/deaths.
- **Universal Basic Income (UBI):** AI may generate enough economic benefit to afford UBI. But studies show **work provides meaning/purpose** → affects well-being.
- **Creative activities:** Perhaps people could work on more creative activities?
- **Cognitive decline:** As humans depend on AI for tasks, they exercise cognitive skills less → skills may decline (e.g., GPS reduces map-reading ability).

9.2.9 Conscious AI and Human Relations

Key Points(If AIs Become Conscious...).

- They should be regarded as **agents with moral status** and awarded rights.
- **Time dilation:** An AI that suffers for 1 minute of human time may suffer for far longer in AI subjective time.

- **How would we know?** Information-theoretic indicators (IIT) + functionalist intuitions (if it behaves as if conscious, it is conscious).

Anthropomorphization

- Humans (especially children) require **very few behavioral cues** to anthropomorphize.
- Strong **commercial motivation** to make AI robots more human-like (speech, gestures, expressions).
- Example: Elderly care robots will be more accepted if they show care, concern, empathy.
- Example: Sex robots will be more desirable if more human-like.

Effects on Human Relationships

#	Concern
1	**Slavery:** Treating conscious robots as slaves (servants, caregivers) might erode moral objection to treating *humans* as slaves (means to ends).
2	**Objectification:** Treating conscious sex robots as objects of gratification might make us more ready to objectify other humans (cf. internet porn's effects on relationships).
3	**Outsourcing care:** If we outsource care to robots, will we become less caring? Will we preserve values of caring for elderly parents, disabled relatives? (Cf. cognitive decline when AI takes over tasks.)
4	**Children and robot companions:** Children test social relationships with dolls. With lifelike robot dolls, they may lose ability to see world through others' eyes → loss of **empathy and responsibility**. ("You dictate relationships, you don't navigate them.")

9.2.10 AI for Improving Human Ethics

Key Points(Positive Possibilities).

- **Joint human-AI reasoning:** For novel ethical problems with no precedent, AI could help humans decide what is right/wrong by providing superior epistemic and causal reasoning, integrated with human input on values/preferences.
- **Cooperative Reinforcement Learning:** AI learns human values from observation; humans teach, AI questions, both learn and act together.
- **AI as empathy machine:** VR could serve as the ultimate "empathy machine"—allowing people to experience the world as others do, widening the circle of concern.

Exam Tip(The Novel and the VR).

Novels have been characterized as "empathy machines"—they help us see/understand/feel the way others do.

AI virtual reality could be even more powerful: immersive experiences that expand our capacity for empathy and help make humans more ethically responsible to others.

9.2.11 Summary: Key Takeaways

Key Points(Week 9 Key Points).

1. **Filter bubbles:** Intellectual isolation from personalized algorithmic filtering.
2. **Echo chambers:** Reinforcement from like-minded social groups.
3. Both lead to: rigidity, entrenchment, polarization, fragmentation of worldviews.
4. **Infopocalypse:** Fake news + deep fakes + filtering + echo chambers → post-truth world.
5. **Argumentative Theory of Reasoning:** Reasoning evolved for social/dialogical contexts, not individual decision-making. Explains confirmation bias and reason-based choice.
6. **Dialogical scaffolding:** Groups reason better; AI can support rational debate and education.
7. **Surveillance capitalism:** Commodification of private experience; behavioral predictions bought/sold for profit.
8. **Attention economy:** Persuasion technologies keep us online; exploit desires for validation.
9. **Harms:** Privacy, political manipulation, loss of serendipity, undermining autonomy, exploiting vulnerability.
10. **AI amplifies** evolved tendencies (belonging, confirmation, acquisition) in contexts where they harm well-being.
11. **AI and work:** Mixed predictions; long-term likely net job loss; ethical issues include meaning/purpose and cognitive decline.
12. **Conscious AI:** Moral status, anthropomorphization, effects on human relationships (slavery, objectification, outsourcing care, children).
13. **AI for good:** Joint reasoning, empathy machines, deliberative democracy.

Definition(Key Terms).

- **Filter bubble:** Algorithmic isolation from diverse views.
- **Echo chamber:** Social reinforcement of like-minded views.
- **Confirmation bias:** Seeking evidence that confirms existing beliefs.
- **Reason-based choice:** Decisions justified by easily defensible reasons.
- **Deep fakes:** AI-generated synthetic media (images, videos).
- **Infopocalypse:** Information apocalypse from fake news and manipulation.
- **Surveillance capitalism:** Commodification of behavioral data for profit.
- **Attention economy:** Competition for human attention via persuasion technologies.
- **Epistemic vigilance:** Evaluating trustworthiness of received information.
- **Anthropomorphization:** Attributing human characteristics to non-humans.
- **UBI:** Universal Basic Income.